



US012418856B2

(12) **United States Patent**
Garcia Azorero et al.

(10) **Patent No.:** **US 12,418,856 B2**

(45) **Date of Patent:** **Sep. 16, 2025**

(54) **PARTIAL SUPPORT OF ACCESS NETWORK INFORMATION**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 427 days.

(21) Appl. No.: **17/800,054**

(22) PCT Filed: **Feb. 17, 2021**

(86) PCT No.: **PCT/IB2021/051354**

§ 371 (c)(1),

(2) Date: **Aug. 16, 2022**

(87) PCT Pub. No.: **WO2021/165856**

PCT Pub. Date: **Aug. 26, 2021**

(65) **Prior Publication Data**

US 2023/0146343 A1 May 11, 2023

Related U.S. Application Data

(60) Provisional application No. 62/977,635, filed on Feb. 17, 2020.

(51) **Int. Cl.**

H04W 48/18 (2009.01)

H04W 4/02 (2018.01)

(52) **U.S. Cl.**

CPC **H04W 48/18** (2013.01); **H04W 4/02** (2013.01)

(58) **Field of Classification Search**

CPC **H04W 48/18**; **H04W 4/02**; **H04L 67/52**; **H04L 65/1016**

USPC **455/435.2**

See application file for complete search history.

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3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; 5G System; Session Management Policy Control Service; Stage 3, 650 Route des Lucioles—Sophia Antipolis (Year: 2019).*

(Continued)

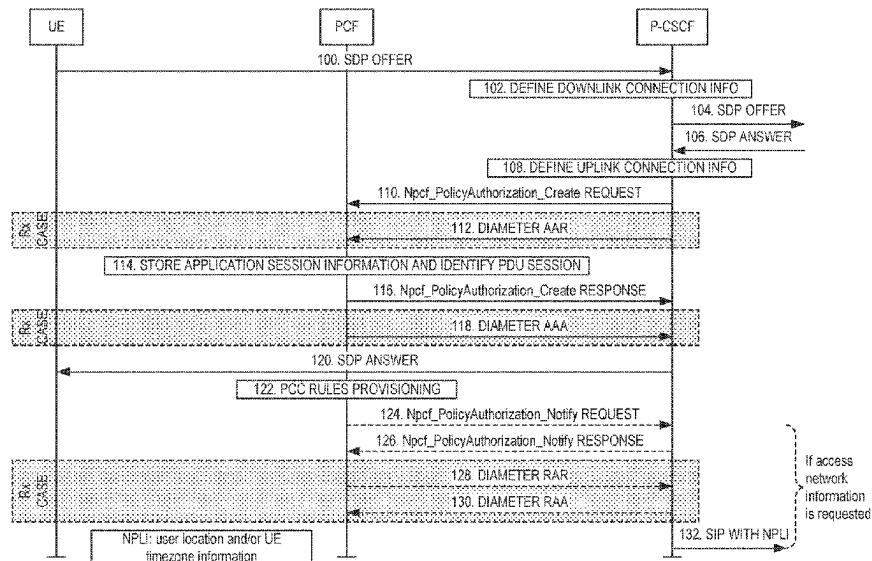
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Intellectual Property Law; Sarvjit S. Patil

(57) **ABSTRACT**

Embodiments of a method performed by a Session Management Function (SMF) for providing Access Network (AN) are disclosed. In some embodiments the method comprises receiving from a Policy Control Function (PCF) a request for AN information. The method also includes determining whether the AN does not support reporting of the requested AN information. The method also includes upon determining that the AN does not support reporting of the requested AN information, sending, to the PCF, a notification that the AN does not support reporting of the requested AN information. The method also includes upon determining that the AN does support reporting of the requested AN information: acquiring the requested AN information and sending the requested AN information to the PCF.

19 Claims, 40 Drawing Sheets



(56)

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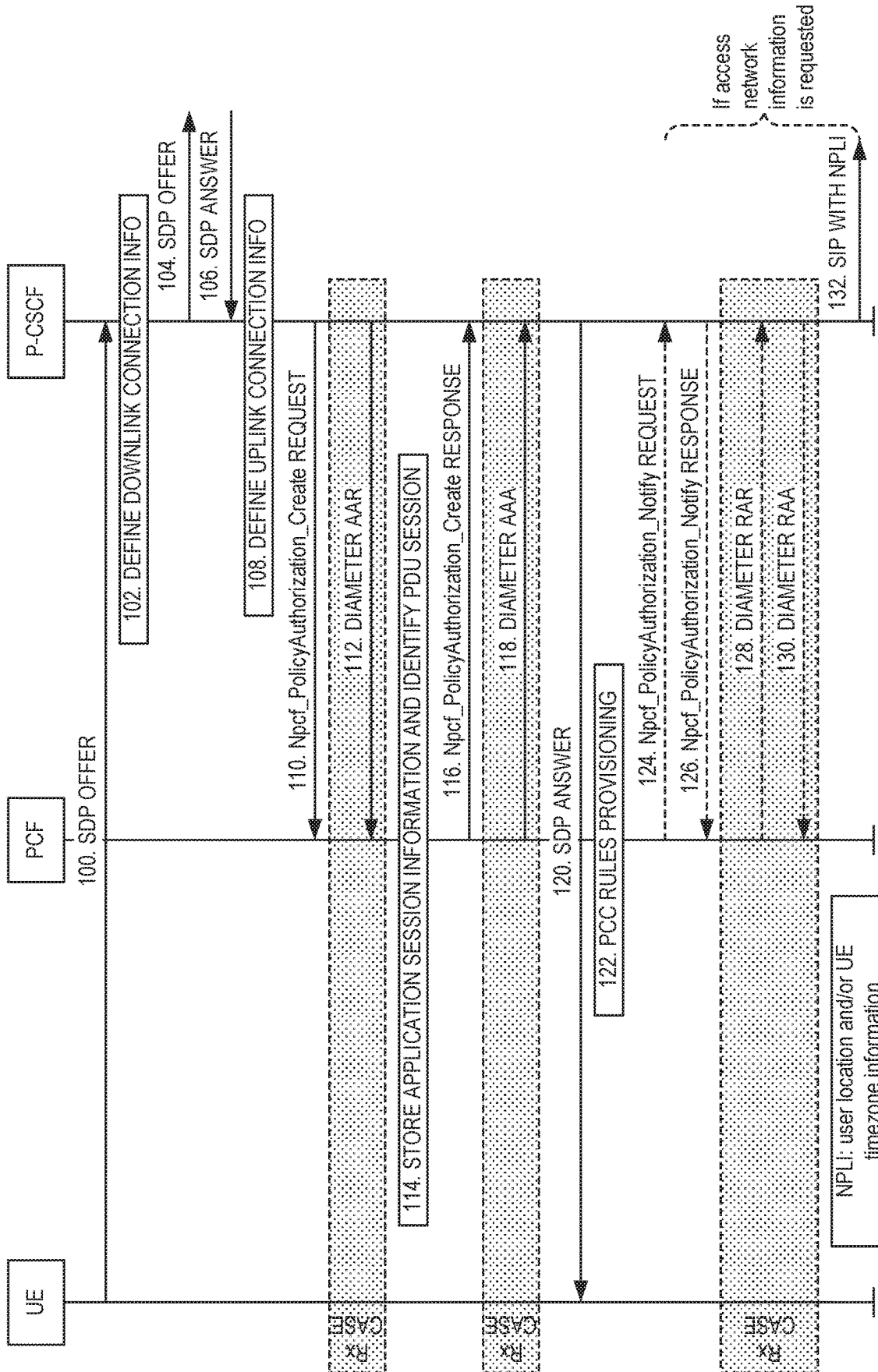


FIG. 1

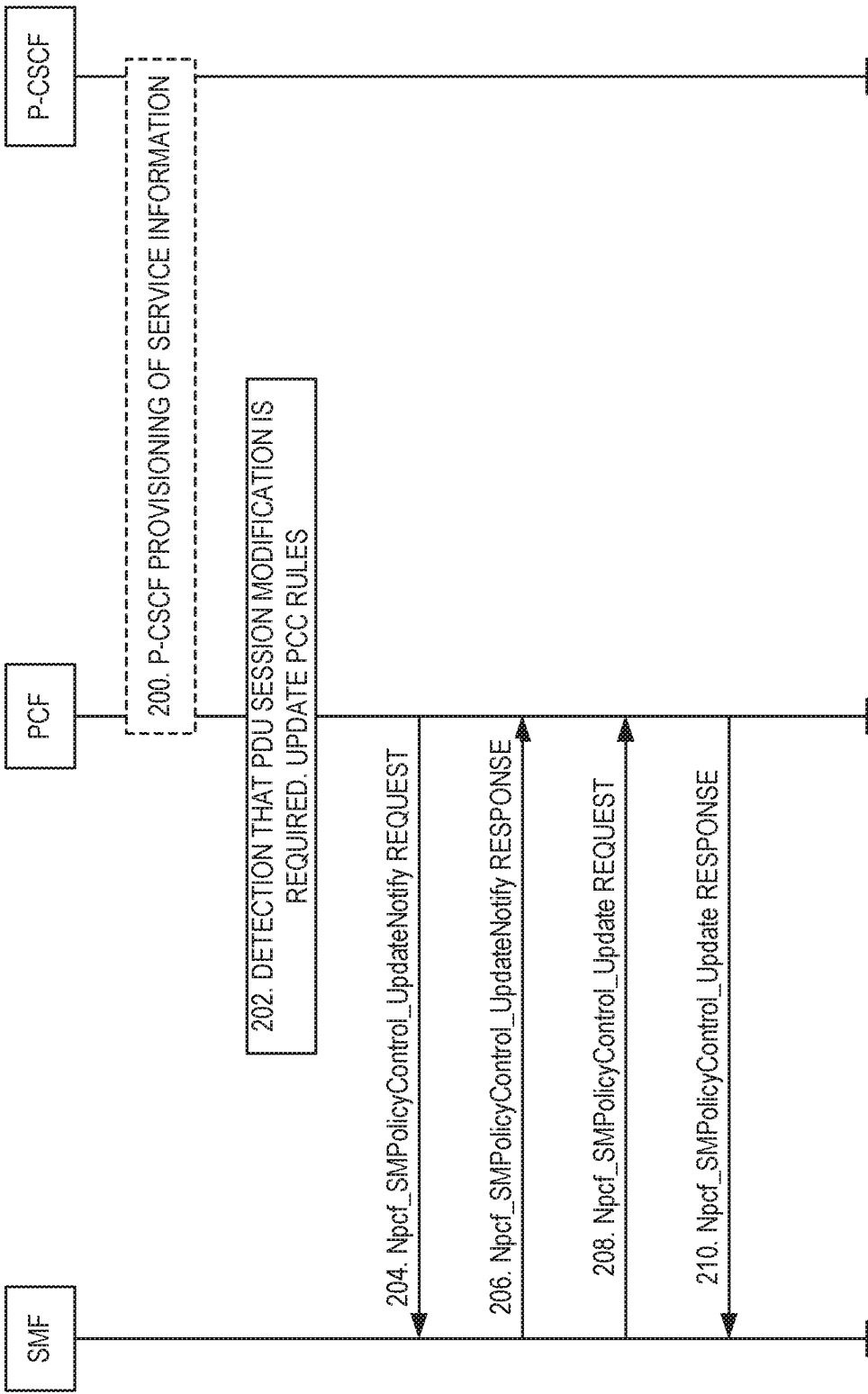


FIG. 2

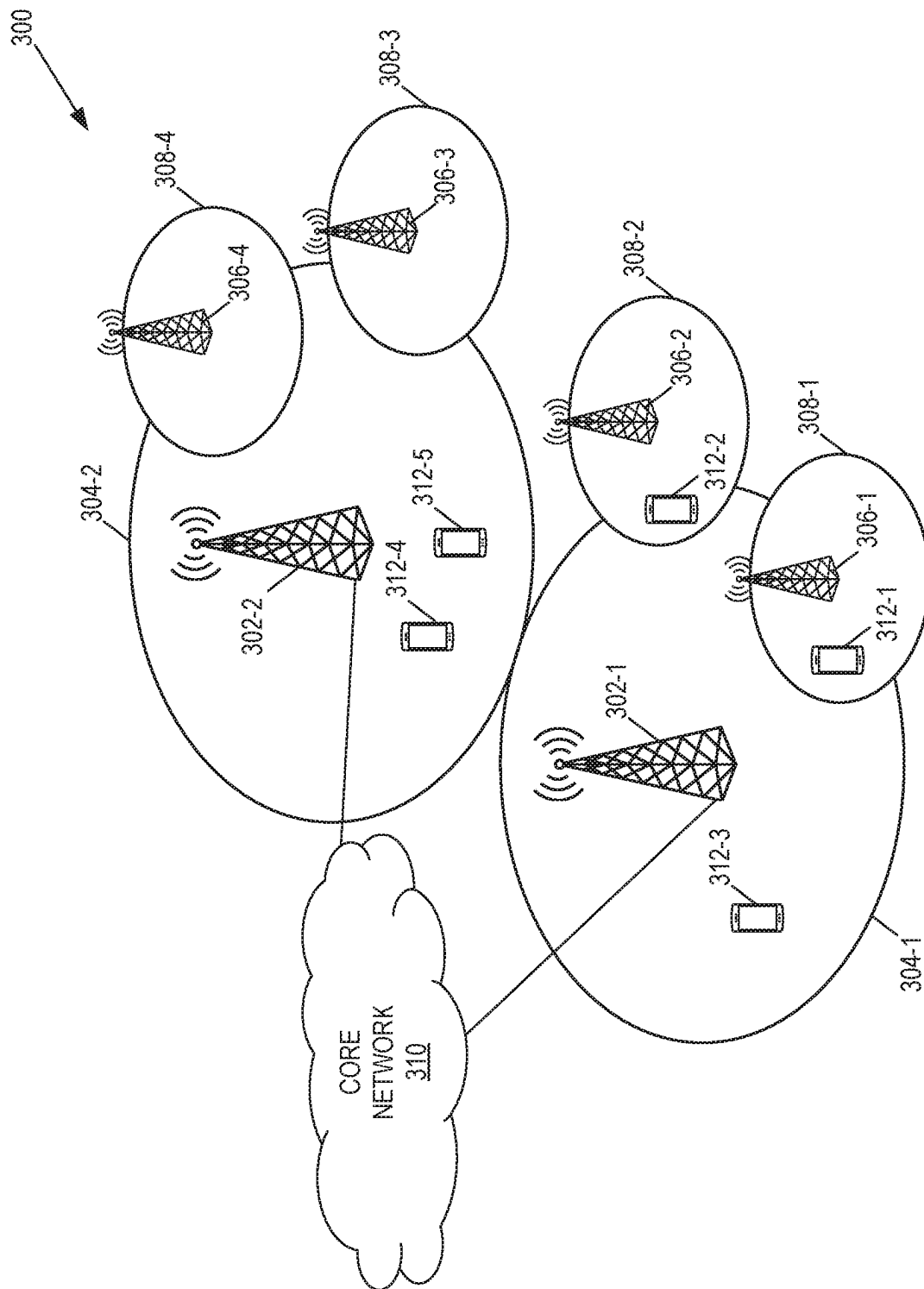


FIG. 3

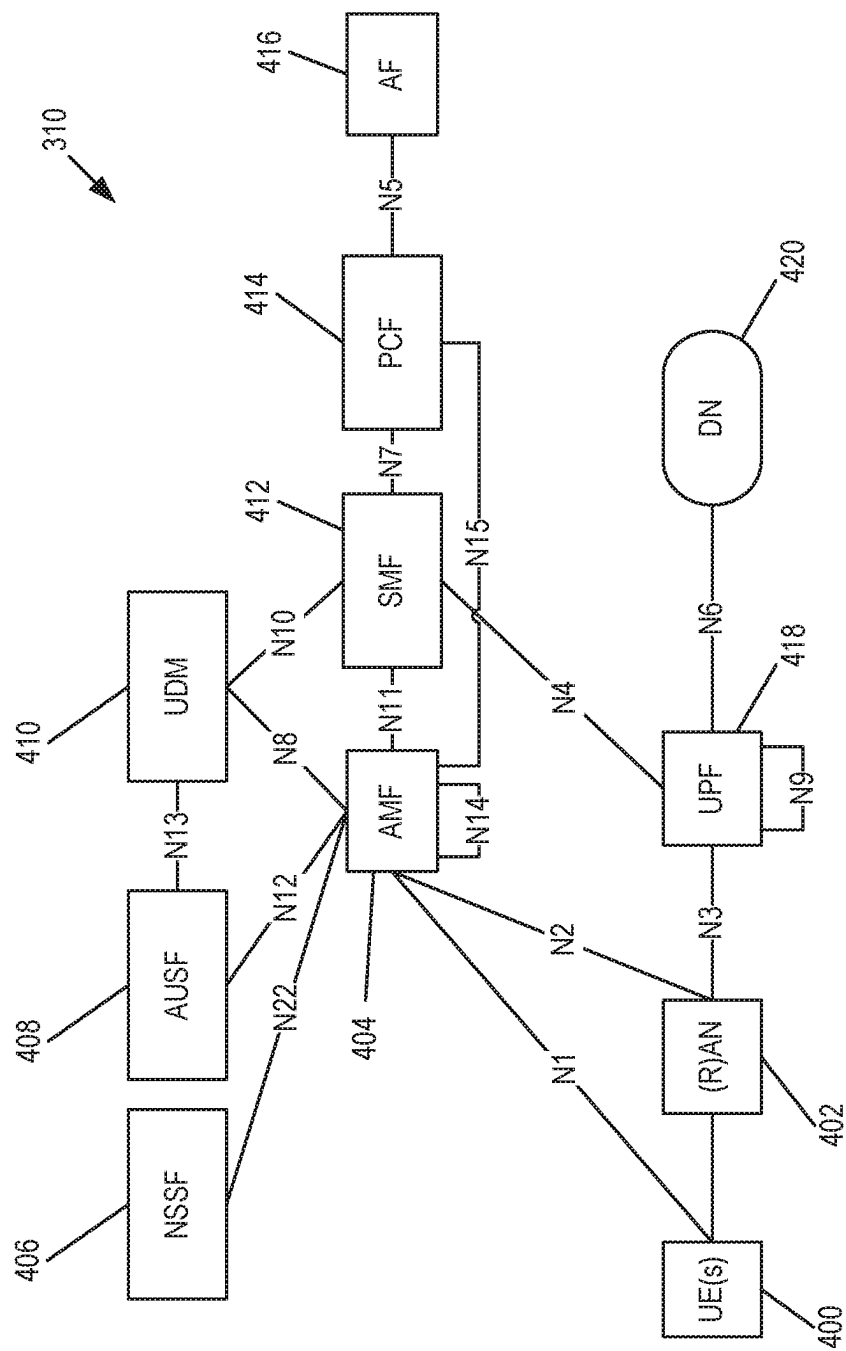


FIG. 4

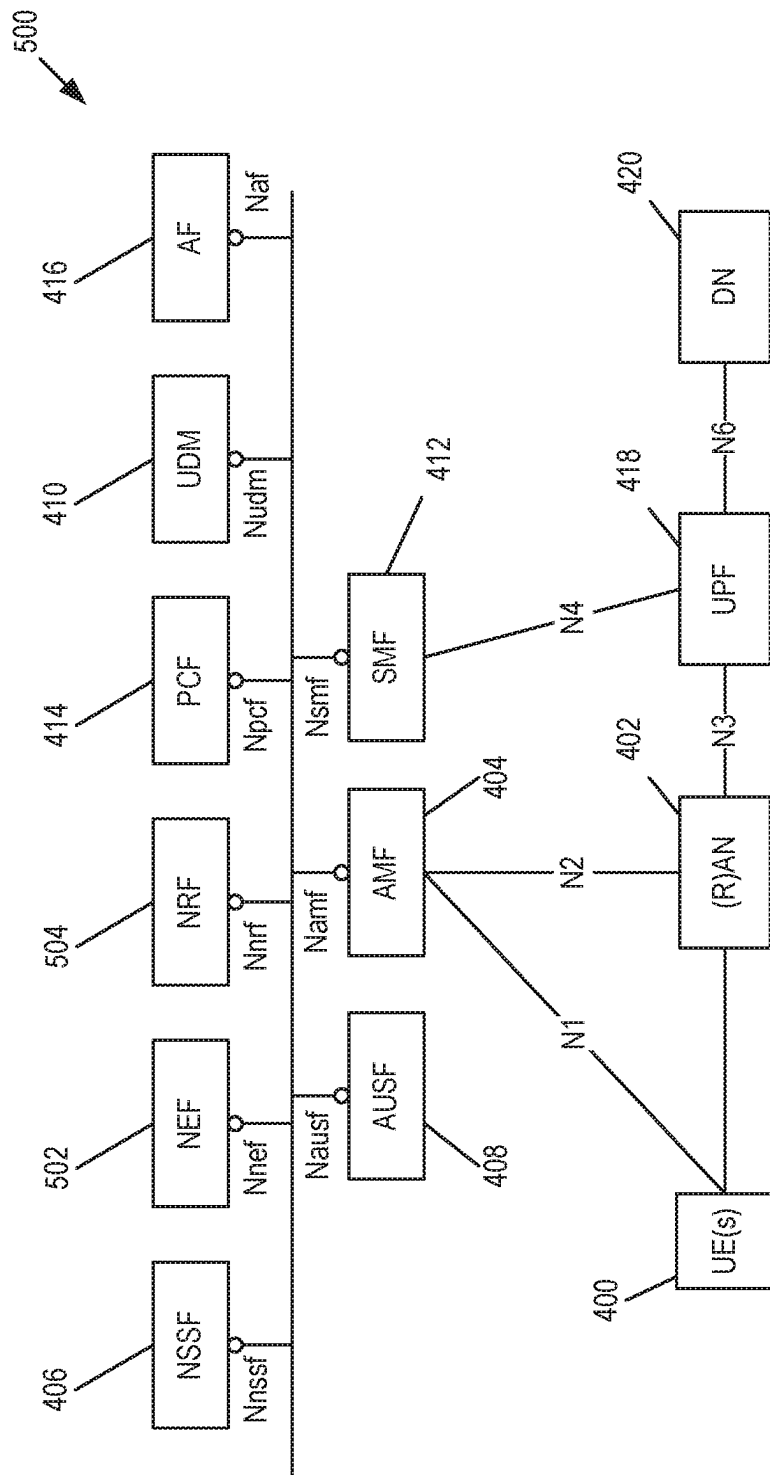


FIG. 5

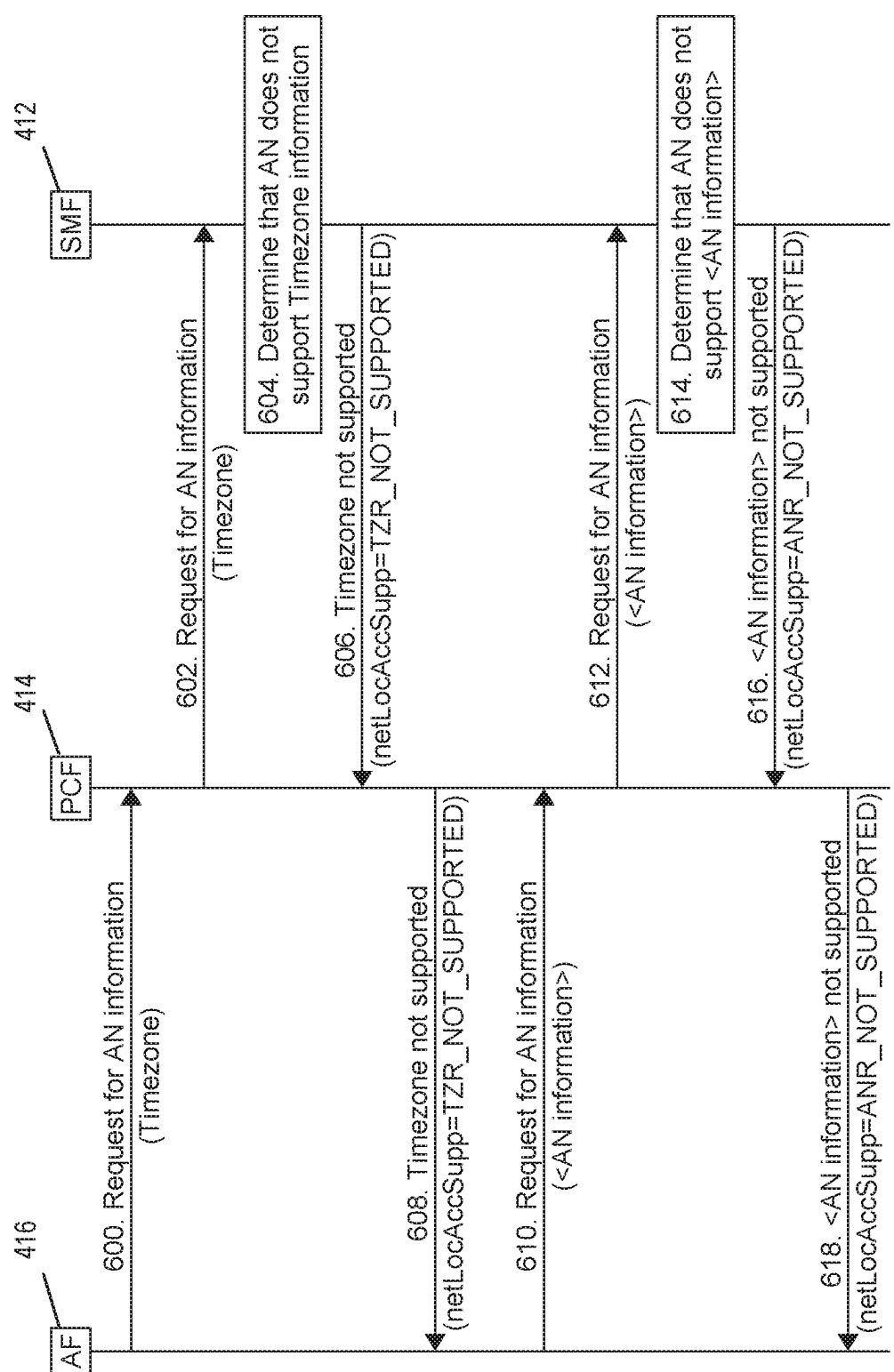


FIG. 6

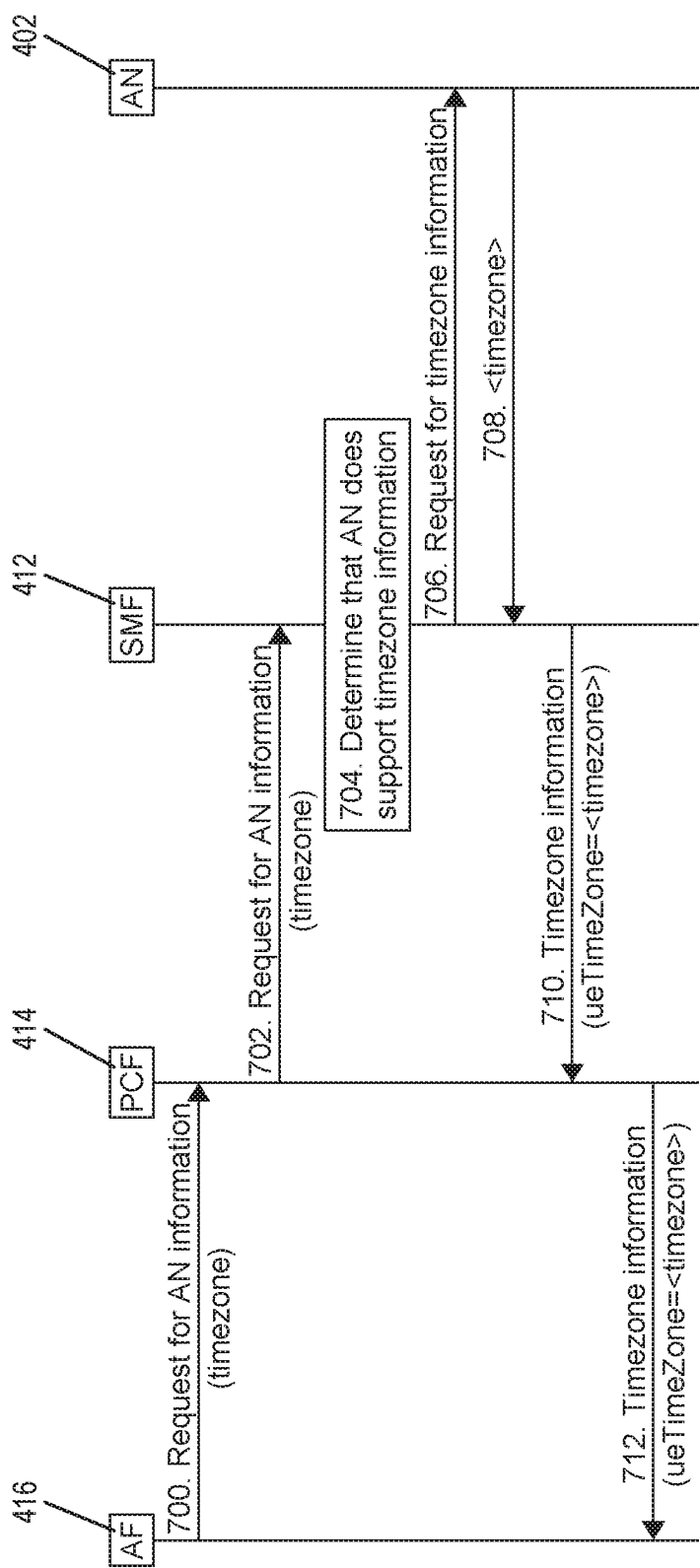


FIG. 7

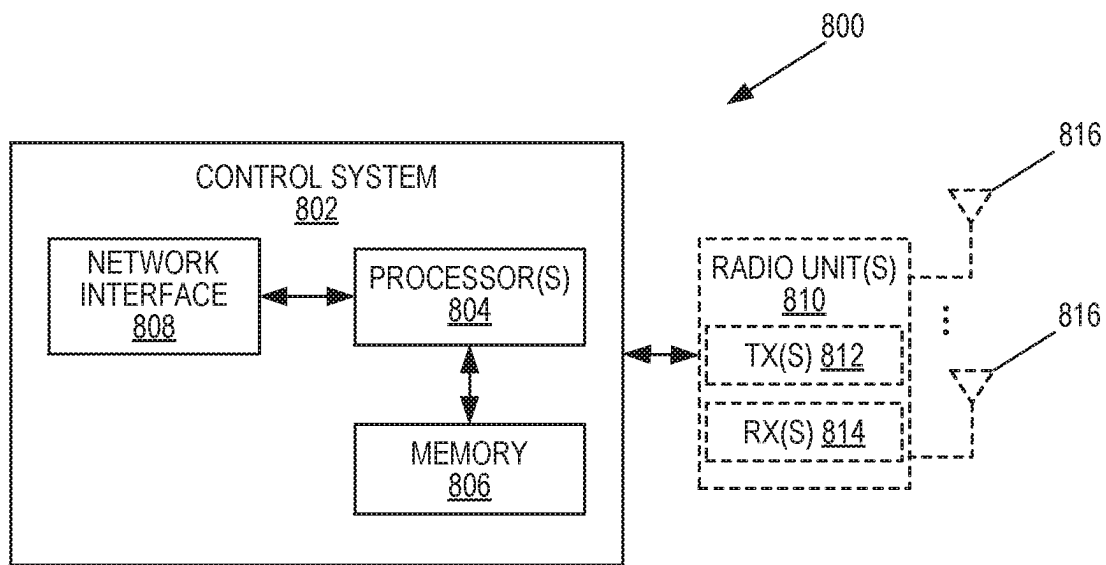


FIG. 8

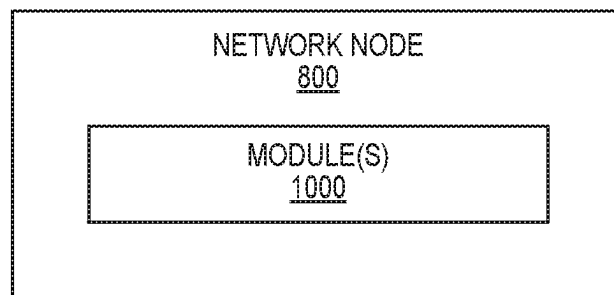
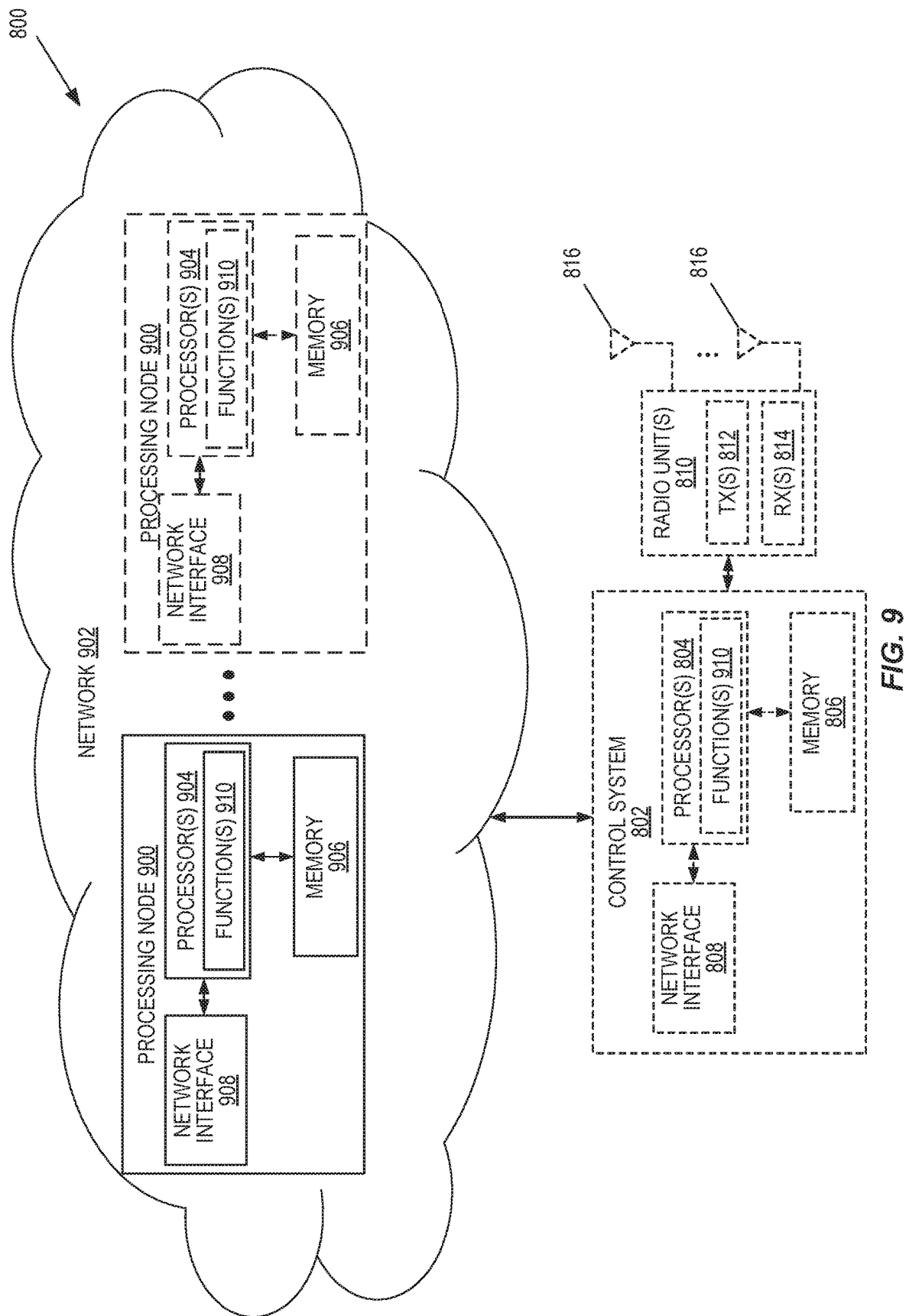
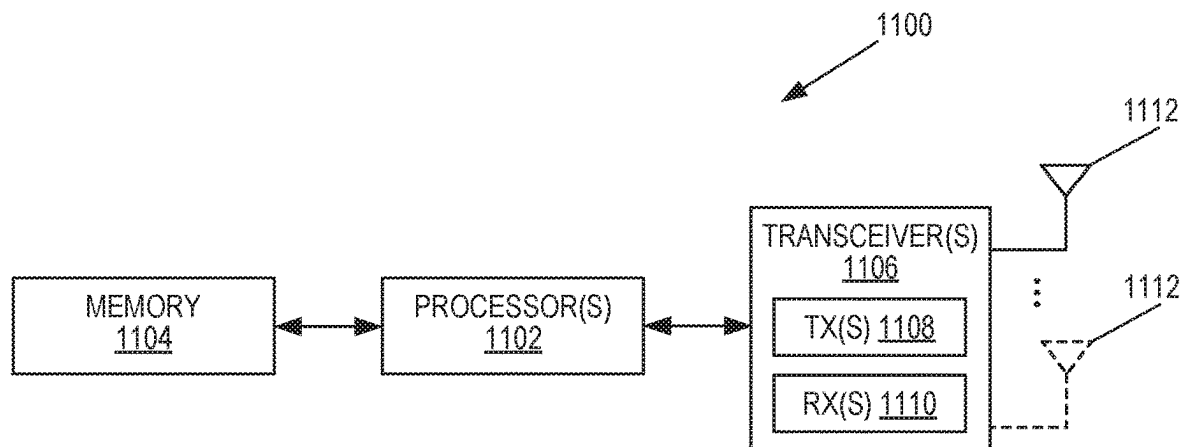
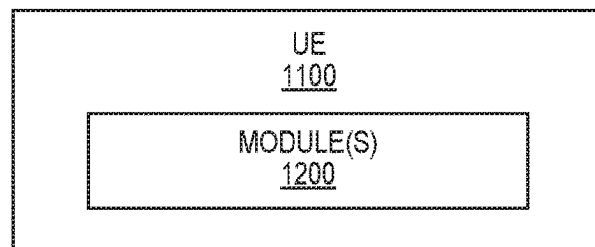


FIG. 10



**FIG. 11****FIG. 12**

***** Start of First Change to 3GPP 29.512 *****

*** 1st Change ***

5.6.1 General

- 5 This subclause specifies the application data model supported by the API.
- The Npcf_SMPolicyControl API allows the SMF to retrieve the session management related policy from the PCF as defined in 3GPP TS 23.503 [6].
- Table 5.6.1-1 specifies the data types defined for the Npcf_SMPolicyControl service based interface protocol.

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Table 5.6.1-1: Npcf_SMPolicyControl specific Data Types

Data type	Section defined	Description	Applicability
5GSmCause	5.6.3.2	Indicates the 5GSM cause code value.	RAN-NAS-Cause
AccNetChargingAddress	5.6.2.35	Identifies the address of the network node performing charging and used for charging applications.	
AccNetChId	5.6.2.23	Contains the access network charging identifier for the PCC rule(s) or whole PDU session.	
AccuUsageReport	5.6.2.18	Contains the accumulated usage report information.	UMC
AfSigProtocol	5.6.3.10	Indicates the protocol used for signalling between the UE and the AF	ProvAFsignalFlow
AppDetectionInfo	5.6.2.22	Contains the detected application's traffic information.	ADC
AuthorizedDefaultQos	5.6.2.34	Authorized Default QoS	
ChargingData	5.6.2.11	Contains charging related parameters.	
ChargingInformation	5.6.2.17	Contains the addresses of the charging functions.	
ConditionData	5.6.2.9	Contains conditions for applicability of a rule.	
CreditManagementStatus	5.6.3.16	Indicates the reason of the credit management session failure.	
ErrorReport	5.6.2.36	Contains the rule reports.	
FailureCause	5.6.3.14	Indicates the cause of the failure in a Partial Success Report.	
FailureCode	5.6.3.9	Indicates the reason of the PCC rule failure.	
FlowDescription	5.6.3.2	Defines a packet filter for an IP flow.	
FlowDirection	5.6.3.3	Indicates the direction of the service data flow.	
FlowDirectionRm	5.6.3.15	This data type is defined in the same way as the "FlowDirection" data type, but with the OpenAPI "nullable: true" property.	
FlowInformation	5.6.2.14	Contains the flow information.	
MeteringMethod	5.6.3.5	Indicates the metering method.	
NetLocAccessSupport	5.6.3.x1	Indicates the access network support of the report of the requested access network information.	NetLoc
PacketFilterContent	5.6.3.2	Defines a packet filter for an IP flow.	
PacketFilterInfo	5.6.2.30	Contains the information from a single packet filter sent from the SMF to the PCF.	
PartialSuccessReport	5.6.2.33	Includes the information reported by the SMF when some of the PCC rules are not successfully installed/activated.	
PccRule	5.6.2.6	Contains the PCC rule information.	

(Continued in Fig. 13B)

Fig. 13A

(Continued from Fig. 13A)

PolicyControlRequestTrigger	5.6.3.6	Contains the policy control request trigger(s).	
QosCharacteristics	5.6.2.16	Contains QoS characteristics for a non-standardized or non-configured 5QI.	
QosData	5.6.2.8	Contains the QoS parameters.	
QosFlowUsage	5.6.3.13	Indicates a QoS flow usage information.	
QosNotificationControlInfo	5.6.2.32	Contains the QoS Notification Control Information.	
RanNasRelCause	5.6.2.28	Contains the RAN/NAS release cause.	RAN-NAS-Cause
RedirectAddressType	5.6.3.12	Indicates the redirect address type.	
RedirectInformation	5.6.2.13	Contains the redirect information.	
ReportingLevel	5.6.3.4	Indicates the reporting level.	
RequestedQos	5.6.2.31	Contains the QoS information requested by the UE	
RequestedRuleData	5.6.2.24	Contains rule data requested by the PCF to receive information associated with PCC rules.	
RequestedRuleDataType	5.6.3.7	Contains the type of rule data requested by the PCF.	
RequestedUsageData	5.6.2.25	Contains usage data requested by the PCF requesting usage reports for the corresponding usage monitoring data instances.	
RuleOperation	5.6.3.11	Indicates a UE initiated resource operation that causes a request for PCC rules.	
RuleReport	5.6.2.27	Reports the status of PCC.	
RuleStatus	5.6.3.8	Indicates the status of PCC or session rule.	
ServingNfIdentity	5.6.2.38	Contains the serving Network Function identity.	
SessionRule	5.6.2.7	Contains session level policy information.	
SessionRuleFailureCode	5.6.3.17	Indicates the reason of the session rule failure.	
SessionRuleReport	5.6.2.37	Reports the status of session rule.	
SmPolicyControl	5.6.2.2	Contains the parameters to request the SM policies and the SM policies authorized by the PCF.	
SmPolicyContextData	5.6.2.3	Contains the parameters to create individual SM policy resource.	
SmPolicyDecision	5.6.2.4	Contains the SM policies authorized by the PCF.	
SmPolicyNotification	5.6.2.5	Contains the update of the SM policies	
SmPolicyDeleteData	5.6.2.15	Contains the parameters to be sent to the PCF when the individual SM policy is deleted.	
SmPolicyUpdateContextData	5.6.2.19	Contains the met policy control request trigger(s) and corresponding new value(s) or the error report of the policy enforcement.	
TerminationNotification	5.6.2.21	Termination Notification	
TrafficControlData	5.6.2.10	Contains parameters determining how flows associated with a PCCRule are treated (blocked, redirected, etc).	
UeCampingRep	5.6.2.26	Contains the current applicable values corresponding to the policy control request triggers.	
UeInitiatedResourceRequest	5.6.2.29	Indicates a UE requests specific QoS handling for selected SDF.	
UpPathChgEvent	5.6.2.20	Contains the UP path change event subscription from the AF	TSC
UsageMonitoringData	5.6.2.12	Contains usage monitoring related control information.	UMC

Table 5.6.1-2 specifies data types re-used by the Npcf_SMPolicyControl service based interface protocol from other specifications, including a reference to their respective specifications and when needed, a short description of their use within the Npcf_SMPolicyControl service based interface.

5

Table 5.6.1-2: Npcf_SMPolicyControl re-used Data Types

Data type	Reference	Comments	Applicability
5GMMCause	3GPP TS 29.571 [11]	Contains the cause value of 5GMM protocol.	RAN-NAS-Cause
5QI	3GPP TS 29.571 [11]	Unsigned integer representing a 5G QoS Identifier (see subclause 5.7.2.1 of 3GPP TS 23.501 [2]), within the range 0 to 255.	
5QIPriorityLevel	3GPP TS 29.571 [11]	Unsigned integer indicating the 5QI Priority Level (see subclauses 5.7.3.3 and 5.7.4 of 3GPP TS 23.501 [2]), within the range 1 to 127. Values are ordered in decreasing order of priority, i.e. with 1 as the highest priority and 127 as the lowest priority.	
5QIPriorityLevelRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "5QIPriorityLevel" data type, but with the OpenAPI "nullable: true" property.	
AccessType	3GPP TS 29.571 [11]	The identification of the type of access network.	
Ambr	3GPP TS 29.571 [11]	Session AMBR	
AnGwAddress	3GPP TS 29.514 [17]	Carries the control plane address of the access network gateway.(NOTE)	
ApplicationChargingId	3GPP TS 29.571 [11]	Application provided charging identifier allowing correlation of charging information	AF_Charging_Identifier
Arp	3GPP TS 29.571 [11]	ARP	
AverWindow	3GPP TS 29.571 [11]	Averaging Window	
AverWindowRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "AverWindow" data type, but with the OpenAPI "nullable: true" property.	
BitRate	3GPP TS 29.571 [11]	String representing a bit rate that shall be formatted as follows: pattern: "^d+(\.d+)?(bps Kbps Mbps Gbps Tbps)\$" Examples: "125 Mbps", "0.125 Gbps", "125000 Kbps"	
BitRateRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "BitRate" data type, but with the OpenAPI "nullable: true" property.	
ChargingId	3GPP TS 29.571 [11]	Charging identifier allowing correlation of charging information	
ContentVersion	3GPP TS 29.514 [17]	Indicates the content version of a PCC rule. It uniquely identifies a version of the PCC rule as defined in subclause 4.2.6.2.14.	RuleVersioning

(Continued in Fig. 13D)

Fig. 13C

(Continued from Fig. 13C)

DateTime	3GPP TS 29.571 [11]	String with format "date-time" as defined in OpenAPI Specification [10].	
DateTimeRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "DateTime" data type, but with the OpenAPI "nullable: true" property.	
DnaiChangeType	3GPP TS 29.571 [11]	Describes the types of DNAI change.	
Dnn	3GPP TS 29.571 [11]	The DNN the user is connected to.	
DurationSec	3GPP TS 29.571 [11]	Identifies a period of time in units of seconds.	
DurationSecRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "DurationSec" data type, but with the OpenAPI "nullable: true" property.	
EthFlowDescription	3GPP TS 29.514 [17]	Defines a packet filter for an Ethernet flow.	
FinalUnitAction	3GPP TS 32.291 [19]	Indicates the action to be taken when the user's account cannot cover the service cost.	
FlowStatus	3GPP TS 29.514 [17]	Describes whether the IP flow(s) are enabled or disabled. The value "REMOVED" is not applicable to Npcf_SMPolicyControl service.	
Gpsi	3GPP TS 29.571 [11]	Identifies a GPSI.	
GroupId	3GPP TS 29.571 [11]	Identifies a group of internal globally unique ID.	
Guami	3GPP TS 29.571 [11]	Globally Unique AMF Identifier.	
IplIndex	3GPP TS 29.519 [15]	Information that identifies which IP pool or external server is used to allocate the IP address.	
Ipv4Addr	3GPP TS 29.571 [11]	Identifies an Ipv4 address.	
Ipv6Addr	3GPP TS 29.571 [11]	Identifies an IPv6 address.	
Ipv6Prefix	3GPP TS 29.571 [11]	The Ipv6 prefix allocated for the user.	
MacAddr48	3GPP TS 29.571 [11]	MAC Address.	
MaxDataBurstVol	3GPP TS 29.571 [11]	Maximum Data Burst Volume	
MaxDataBurstVolRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "MaxDataBurstVol" data type, but with the OpenAPI "nullable: true" property.	
NetworkId	3GPP TS 29.571 [11]	The identification of the Network.	
NfInstanceId	3GPP TS 29.571 [11]	The NF instance identifier.	
NgApCause	3GPP TS 29.571 [11]	Contains the cause value of NgAP protocol.	RAN-NAS-Cause
PacketDelBudget	3GPP TS 29.571 [11]	Packet Delay Budget	
PacketErrRate	3GPP TS 29.571 [11]	Packet Error Rate	
PacketLossRateRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "PacketLossRate" data type, but with the OpenAPI "nullable: true" property.	
PduSessionId	3GPP TS 29.571 [11]	The identification of the PDU session.	
PduSessionType	3GPP TS 29.571 [11]	Indicate the type of a PDU session	
Pei	3GPP TS 29.571 [11]	The Identification of a Permanent Equipment.	
PolicyAssociationReleaseCause	3GPP TS 29.507 [25]	The cause why the PCF requests the termination of the policy association	
PresenceInfo	3GPP TS 29.571 [11]	Contains the information which describes a Presence Reporting Area.	PRA
PresenceInfoRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "PresenceInfo" data type, but with the OpenAPI "nullable: true" property.	
ProblemDetails	3GPP TS 29.571 [11]	Contains a detailed information about an error.	
QosNotifType	3GPP TS 29.514 [17]	Indicates whether the GBR targets for the indicated SDFs are "NOT_GUARANTEED" or "GUARANTEED" again.	
QosResourceType	3GPP TS 29.571 [11]	Indicates whether the resource type is GBR, delay critical GBR, or non-GBR.	
RatingGroup	3GPP TS 29.571 [11]	Identifier of a rating group	
RatType	3GPP TS 29.571 [11]	The identification of the RAT type.	
RouteToLocation	3GPP TS 29.571 [11]	A traffic routes to applications location.	TSC
Serviceld	3GPP TS 29.571 [11]	Identifier of a service	

(Continued in Fig. 13E)

(Continued from Fig. 13D)

Snssai	3GPP TS 29.571 [11]	Identifies the S-NSSAI.	
SubscribedDefaultQos	3GPP TS 29.571 [11]	Subscribed Default QoS.	
Supi	3GPP TS 29.571 [11]	The identification of the user (i.e. IMSI, NAI).	
SupportedFeatures	3GPP TS 29.571 [11]	Used to negotiate the applicability of the optional features defined in table 5.8-1.	
TraceData	3GPP TS 29.571 [11]		
TimeZone	3GPP TS 29.571 [11]	Contains the user time zone information.	
Uri	3GPP TS 29.571 [11]	URI.	
UserLocation	3GPP TS 29.571 [11]	Contains the user location	
Volume	3GPP TS 29.122 [32]	Unsigned integer identifying a volume in units of bytes.	
VolumeRm	3GPP TS 29.122 [32]	This data type is defined in the same way as the "VolumeRm" data type, but with the OpenAPI "nullable: true" property.	
NOTE: "AnGwAddr" data structure is only applicable to the 5GS and EPC/E-UTRAN interworking scenario as defined in Annex B.			

*** 2nd Change ***

5

5.6.2.26 Type UeCampingRep**Table 5.6.2.26-1: Definition of type UeCampingRep**

Attribute name	Data type	P	Cardinality	Description	Applicability
accessType	AccessType	O	0..1	The Access Type where the served UE is camping.	
ratType	RatType	O	0..1	The RAT Type where the served UE is camping.	
servNfId	ServingNfIdentity	O	0..1	Contains the serving network function identity.	
servingNetwork	NetworkId	O	0..1	The serving network where the served UE is camping.	
userLocationInfo	UserLocation	O	0..1	The location of the served UE is camping.	
ueTimeZone	TimeZone	O	0..1	The time zone where the served UE is camping.	
netLocAccSupp	NetLocAccessSupport	O	0..1	Indicates the access network does not support the report of the requested access network information. The absence of this attribute indicates that the access network supports access network information reporting.	NetLoc

10

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Fig. 13E

*** 3rd Change ***

5.6.3.x1 Enumeration: NetLocAccessSupport

Table 5.6.3.x1-1: Enumeration NetLocAccessSupport

Enumeration value	Description	Applicability
ANR_NOT_SUPPORTED	Indicates that the access network does not support the report of access network information.	
TZR_NOT_SUPPORTED	Indicates that the access network does not support the report of UE time zone. (NOTE)	
NOTE: The UE time zone is not available in EPC untrusted WLAN.		

*** 4th Change ***

A.2 Npcf_SMPolicyControl API

```

openapi: 3.0.0
info:
  title: Npcf_SMPolicyControl API
  version: 1.0.4
  description: |
    Session Management Policy Control Service
    © 2019, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.
externalDocs:
  description: 3GPP TS 29.512 V15.6.0; 5G System; Session Management Policy Control Service.
  url: 'http://www.3gpp.org/ftp/Specs/archive/29_series/29.512/'
:
components:
  :
  schemas:
    :
    UeCampingRep:
      type: object
      properties:
        accessType:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
        ratType:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/RatType'
        servNfId:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/ServingNfIdentity'
        servingNetwork:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NetworkId'
        userLocationInfo:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/UserLocation'
        ueTimeZone:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/TimeZone'
        netLocAccSupp:
          $ref: 'TS29571_CommonData.yaml#/components/schemas/NetLocAccessSupport'
  :
  NetLocAccessSupport:
    anyOf:
      - type: string
      enum:
        - ANR_NOT_SUPPORTED
        - TZR_NOT_SUPPORTED
      - type: string
      description: >
        This string provides forward-compatibility with future
        extensions to the enumeration but is not used to encode
        content defined in the present version of this API.
      description: >
        Possible values are
        - ANR_NOT_SUPPORTED: Indicates that the access network does not support the report of
        access network information.
        - TZR_NOT_SUPPORTED: Indicates that the access network does not support the report of UE
        time zone.

```

Fig. 13F

*** 5th Change ***

B.3.2.0 General

When the UE establishes the PDN connection through the EPC network and the SMF+PGW-C receives the Create Session Request message as defined in

- 5 3GPP TS 29.274 [37], the SMF+PGW-C shall behave as defined in subclause 4.2.2.2 with the differences that the SMF+PGW-C shall include (if available) in SmPolicyContextData data structure:

- the IMSI of the user within the "supi" attribute;
- the MSISDN of the user within the "gpsi" attribute;
- 10 - APN within the "dnn" attribute;
- PDN Type within the "pduSessionType" attribute;
- IMEI-SV within the "pei" attribute;
- IP-CAN type within the "accessType" attribute;
- RAT type within the "ratType" attribute;

15 NOTE 1: See Annex B.3.2.2 for further information.

- subscribed APN-AMBR within "subsSessAmbr" attribute;
- subscribed Default EPS bearer QoS within "subsDefQos" attribute;

NOTE 2: subscribed APN-AMBR and the QCI within the subscribed default EPS bearer QoS are mapped to subscribed Session-AMBR and 5QI as defined in Annex B.3.6.1 respectively.

- 20 - user location information within the "userLocationInfo" attribute;

NOTE 3: See Annex B.3.2.1 for further information.

- the S-NSSAI determined by the SMF+PGW-C within the "sliceInfo" attribute; and
- the bearer usage required of the default bearer within the "qosFlowUsage" attribute.
- the UE time zone information within "ueTimeZone" attribute, if available;

25 NOTE x1: The UE time zone is not available in EPC untrusted WLAN.

30

Fig. 13G

*** 6th Change ***

B.3.6.x1 Request and Report of Access Network information

If the NetLoc feature as defined in subclause 5.8 is supported, the PCF may request the
5 SMF+PGW-C to report the access network information as defined in
subclause 4.2.6.5.4.

If the AN_INFO policy control request trigger is set, upon receiving the
"lastReqRuleData" attribute with the "reqData" attribute with the value(s)
MS_TIME_ZONE and/or USER_LOC_INFO and the "refPccRuleIds" attribute containing
10 the PCC rule identifier(s) corresponding to the PCC rule(s) being installed, modified or
removed:

- If the SMF+PGW-C determines that the access network does not support the access network information
reporting based on the feature support, the SMF+PGW-C shall immediately inform the PCF by including the
15 "netLocAccSupp" attribute set to "ANR_NOT_SUPPORTED" value in the "UeCampingRep" data structure
returned in the "200 OK" response to the policy update notification request.

- Otherwise:

- a) If the "reqData" attribute includes the MS_TIME_ZONE value and the SMF+PGW-C determines that the
access network does not support the report of the UE time zone, the SMF+PGW-C shall immediately
inform the PCF by including the "netLocAccSupp" attribute set to "TZR_NOT_SUPPORTED" value in
20 the "UeCampingRep" data structure returned in the "200 OK" response to the policy update notification
request.

- b) If the "reqData" attribute includes:

- 1) the USER_LOC_INFO value; and/or

- 2) the MS_TIME_ZONE value and the SMF+PGW-C determines the access network supports the report
25 of UE time zone,

the SMF+PGW-C shall apply appropriate procedures to the EPC access network to obtain the requested
access network information and shall behave as specified in subclause 4.2.4.9.

Fig. 13H

***** Start of First Change to 3GPP 29.512 *****

*** 1st Change ***

5.6.1 General

- 5 This subclause specifies the application data model supported by the API.
- The Npcf_SMPolicyControl API allows the SMF to retrieve the session management related policy from the PCF as defined in 3GPP TS 23.503 [6].
- Table 5.6.1-1 specifies the data types defined for the Npcf_SMPolicyControl service based interface protocol.

10

Table 5.6.1-1: Npcf_SMPolicyControl specific Data Types

Data type	Section defined	Description	Applicability
5GSMCause	5.6.3.2	Indicates the 5GSM cause code value.	RAN-NAS-Cause
AccNetChargingAddress	5.6.2.35	Identifies the address of the network node performing charging and used for charging applications.	
AccNetChId	5.6.2.23	Contains the access network charging identifier for the PCC rule(s) or whole PDU session.	
AccuUsageReport	5.6.2.18	Contains the accumulated usage report information.	UMC
AFSigProtocol	5.6.3.10	Indicates the protocol used for signalling between the UE and the AF	ProvAFsignalFlow
AppDetectionInfo	5.6.2.22	Contains the detected application's traffic information.	ADC
AuthorizedDefaultQos	5.6.2.34	Authorized Default QoS	
ChargingData	5.6.2.11	Contains charging related parameters.	
ChargingInformation	5.6.2.17	Contains the addresses of the charging functions.	
ConditionData	5.6.2.9	Contains conditions for applicability of a rule.	
CreditManagementStatus	5.6.3.16	Indicates the reason of the credit management session failure.	
ErrorReport	5.6.2.36	Contains the rule reports.	
FailureCause	5.6.3.14	Indicates the cause of the failure in a Partial Success Report.	
FailureCode	5.6.3.9	Indicates the reason of the PCC rule failure.	
FlowDescription	5.6.3.2	Defines a packet filter for an IP flow.	
FlowDirection	5.6.3.3	Indicates the direction of the service data flow.	
FlowDirectionRm	5.6.3.15	This data type is defined in the same way as the "FlowDirection" data type, but with the OpenAPI "nullable: true" property.	
FlowInformation	5.6.2.14	Contains the flow information.	
MeteringMethod	5.6.3.5	Indicates the metering method.	
MulticastAccessControl	5.6.3.20	Indicates whether the service data flow, corresponding to the service data flow template, is allowed or not allowed.	WWC
NetLocAccessSupport	5.6.3.x1	Indicates the access network support of the report of the requested access network information.	NetLoc
RequestedQosMonitoringParameter	5.6.3.21	Indicates the requested QoS monitoring parameters to be measured.	QosMonitoring

(Continued in Fig. 14B)

Fig. 14A

(Continued from Fig. 14A)

PacketFilterContent	5.6.3.2	Defines a packet filter for an IP flow.	
PacketFilterInfo	5.6.2.30	Contains the information from a single packet filter sent from the SMF to the PCF.	
PartialSuccessReport	5.6.2.33	Includes the information reported by the SMF when some of the PCC rules are not successfully installed/activated.	
PccRule	5.6.2.6	Contains the PCC rule information.	
PduSessionRelCause	5.6.3.24	Contains the SMF PDU Session release cause.	PDU Session Release Cause
PolicyControlRequestTrigger	5.6.3.6	Contains the policy control request trigger(s).	
QosCharacteristics	5.6.2.16	Contains QoS characteristics for a non-standardized or non-configured 5QI.	
QosData	5.6.2.8	Contains the QoS parameters.	
QosFlowUsage	5.6.3.13	Indicates a QoS flow usage information.	
QosMonitoringData	5.6.2.40	Contains QoS monitoring related control information.	QoS Monitoring
QosMonitoringReport	5.6.2.42	Contains QoS monitoring reporting information.	QoS Monitoring
QosNotificationControlInfo	5.6.2.32	Contains the QoS Notification Control Information.	
RanNasRelCause	5.6.2.28	Contains the RAN/NAS release cause.	RAN-NAS-Cause
RedirectAddressType	5.6.3.12	Indicates the redirect address type.	
RedirectInformation	5.6.2.13	Contains the redirect information.	
ReportingFrequency	5.6.3.22	Indicates the frequency for the reporting	QoS Monitoring
ReportingLevel	5.6.3.4	Indicates the reporting level.	
RequestedQos	5.6.2.31	Contains the QoS information requested by the UE	
RequestedQosMonitoringParameter	5.6.3.21	Indicates the requested QoS monitoring parameters to be measured.	QoS Monitoring
RequestedRuleData	5.6.2.24	Contains rule data requested by the PCF to receive information associated with PCC rules.	
RequestedRuleDataType	5.6.3.7	Contains the type of rule data requested by the PCF.	
RequestedUsageData	5.6.2.25	Contains usage data requested by the PCF requesting usage reports for the corresponding usage monitoring data instances.	
RuleOperation	5.6.3.11	Indicates a UE initiated resource operation that causes a request for PCC rules.	
RuleReport	5.6.2.27	Reports the status of PCC.	
RuleStatus	5.6.3.8	Indicates the status of PCC or session rule.	
ServingNfIdentity	5.6.2.38	Contains the serving Network Function identity.	
SessionRule	5.6.2.7	Contains session level policy information.	
SessionRuleFailureCode	5.6.3.17	Indicates the reason of the session rule failure.	
SessionRuleReport	5.6.2.37	Reports the status of session rule.	
SmPolicyAssociationReleaseCause	5.6.3.23	Represents the cause why the PCF requests the termination of the SM policy association.	
SmPolicyControl	5.6.2.2	Contains the parameters to request the SM policies and the SM policies authorized by the PCF.	
SmPolicyContextData	5.6.2.3	Contains the parameters to create individual SM policy resource.	
SmPolicyDecision	5.6.2.4	Contains the SM policies authorized by the PCF.	
SmPolicyNotification	5.6.2.5	Contains the update of the SM policies	
SmPolicyDeleteData	5.6.2.15	Contains the parameters to be sent to the PCF when the individual SM policy is deleted.	
SmPolicyUpdateContextData	5.6.2.19	Contains the met policy control request trigger(s) and corresponding new value(s) or the error report of the policy enforcement.	

(Continued in Fig. 14C)

(Continued from Fig. 14B)

SteeringFunctionality	5.6.3.18	Indicates functionality to support traffic steering, switching and splitting determined by the PCF.	ATSSS
SteeringMode	5.6.2.39	Contains the steering mode value and parameters determined by the PCF.	ATSSS
SteeringModeValue	5.6.3.19	Indicates the steering mode value determined by the PCF.	ATSSS
TerminationNotification	5.6.2.21	Termination Notification	
TrafficControlData	5.6.2.10	Contains parameters determining how flows associated with a PCCRule are treated (blocked, redirected, etc).	
TsnBridgeInfo	5.6.2.41	Contains parameters that describe and identify the TSN bridge.	TimeSensitive Networking
UeCampingRep	5.6.2.26	Contains the current applicable values corresponding to the policy control request triggers.	
UeInitiatedResourceRequest	5.6.2.29	Indicates a UE requests specific QoS handling for selected SDF.	
UpPathChgEvent	5.6.2.20	Contains the UP path change event subscription from the AF	TSC
UsageMonitoringData	5.6.2.12	Contains usage monitoring related control information.	UMC

Table 5.6.1-2 specifies data types re-used by the Npcf_SMPolicyControl service based interface protocol from other specifications, including a reference to their respective

- 5 specifications and when needed, a short description of their use within the Npcf_SMPolicyControl service based interface.

Table 5.6.1-2: Npcf_SMPolicyControl re-used Data Types

Data type	Reference	Comments	Applicability
5GMMCause	3GPP TS 29.571 [11]	Contains the cause value of 5GMM protocol.	RAN-NAS-Cause
5Qi	3GPP TS 29.571 [11]	Unsigned integer representing a 5G QoS Identifier (see subclause 5.7.2.1 of 3GPP TS 23.501 [2]), within the range 0 to 255.	
5QIPriorityLevel	3GPP TS 29.571 [11]	Unsigned integer indicating the 5QI Priority Level (see subclauses 5.7.3.3 and 5.7.4 of 3GPP TS 23.501 [2]), within the range 1 to 127. Values are ordered in decreasing order of priority, i.e. with 1 as the highest priority and 127 as the lowest priority.	
5QIPriorityLevelRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "5QIPriorityLevel" data type, but with the OpenAPI "nullable: true" property.	
AccessType	3GPP TS 29.571 [11]	The identification of the type of access network.	
Ambr	3GPP TS 29.571 [11]	Session AMBR.	
AnGwAddress	3GPP TS 29.514 [17]	Carries the control plane address of the access network gateway. (NOTE 1)	
ApplicationChargingId	3GPP TS 29.571 [11]	Application provided charging identifier allowing correlation of charging information.	AF_Charging_Identifier
Arp	3GPP TS 29.571 [11]	ARP.	
AverWindow	3GPP TS 29.571 [11]	Averaging Window.	

(Continued in Fig. 14D)

(Continued from Fig. 14C)

AverWindowRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "AverWindow" data type, but with the OpenAPI "nullable: true" property.	
Bytes	3GPP TS 29.571 [11]	String with format "byte".	TimeSensitive Networking
BitRate	3GPP TS 29.571 [11]	String representing a bit rate that shall be formatted as follows: pattern: "^d+(\.d+)? (bps Kbps Mbps Gbps Tbps)\$" Examples: "125 Mbps", "0.125 Gbps", "125000 Kbps".	
BitRateRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "BitRate" data type, but with the OpenAPI "nullable: true" property.	
ChargingId	3GPP TS 29.571 [11]	Charging identifier allowing correlation of charging information.	
ContentVersion	3GPP TS 29.514 [17]	Indicates the content version of a PCC rule. It uniquely identifies a version of the PCC rule as defined in subclause 4.2.6.2.14.	RuleVersionin g
DateTime	3GPP TS 29.571 [11]	String with format "date-time" as defined in OpenAPI Specification [10].	
DateTimeRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "DateTime" data type, but with the OpenAPI "nullable: true" property.	
DnaiChangeType	3GPP TS 29.571 [11]	Describes the types of DNAI change.	
Dnn	3GPP TS 29.571 [11]	The DNN the user is connected to.	
DurationSec	3GPP TS 29.571 [11]	Identifies a period of time in units of seconds.	
DurationSecRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "DurationSec" data type, but with the OpenAPI "nullable: true" property.	
EthFlowDescription	3GPP TS 29.514 [17]	Defines a packet filter for an Ethernet flow. (NOTE 2)	
ExtMaxDataBurstVol	3GPP TS 29.571 [11]	Maximum Data Burst Volume.	EMDBV
ExtMaxDataBurstVolRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "ExtMaxDataBurstVol" data type, but with the OpenAPI "nullable: true" property.	EMDBV
FinalUnitAction	3GPP TS 32.291 [19]	Indicates the action to be taken when the user's account cannot cover the service cost.	
FlowStatus	3GPP TS 29.514 [17]	Describes whether the IP flow(s) are enabled or disabled. The value "REMOVED" is not applicable to Npcf_SMPolicyControl service.	
Gpsi	3GPP TS 29.571 [11]	Identifies a GPSI.	
GroupId	3GPP TS 29.571 [11]	Identifies a group of internal globally unique ID.	
Guami	3GPP TS 29.571 [11]	Globally Unique AMF Identifier.	
IpIndex	3GPP TS 29.519 [15]	Information that identifies which IP pool or external server is used to allocate the IP address.	
Ipv4Addr	3GPP TS 29.571 [11]	Identifies an Ipv4 address.	
Ipv6Addr	3GPP TS 29.571 [11]	Identifies an IPv6 address.	
Ipv6Prefix	3GPP TS 29.571 [11]	The Ipv6 prefix allocated for the user.	
MacAddr48	3GPP TS 29.571 [11]	MAC Address.	
MaxDataBurstVol	3GPP TS 29.571 [11]	Maximum Data Burst Volume.	
MaxDataBurstVolRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "MaxDataBurstVol" data type, but with the OpenAPI "nullable: true" property.	
NfInstanceId	3GPP TS 29.571 [11]	The NF instance identifier.	
NgApCause	3GPP TS 29.571 [11]	Contains the cause value of NgAP protocol.	RAN-NAS- Cause

(Continued in Fig. 14E)

Fig. 14D

(Continued in Fig. 14D)

PacketDelBudget	3GPP TS 29.571 [11]	Packet Delay Budget.	
PacketErrRate	3GPP TS 29.571 [11]	Packet Error Rate.	
PacketLossRateRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "PacketLossRate" data type, but with the OpenAPI "nullable: true" property.	
PduSessionId	3GPP TS 29.571 [11]	The identification of the PDU session.	
PduSessionType	3GPP TS 29.571 [11]	Indicate the type of a PDU session.	
Pei	3GPP TS 29.571 [11]	The Identification of a Permanent Equipment.	
PlmnIdNid	3GPP TS 29.571 [11]	The identification of the Network. PLMN Identity, and for SNPN NID.	
PolicyAssociationReleaseCause	3GPP TS 29.507 [25]	The cause why the PCF requests the termination of the policy association.	
PresenceInfo	3GPP TS 29.571 [11]	Contains the information which describes a Presence Reporting Area.	PRA
PresenceInfoRm	3GPP TS 29.571 [11]	This data type is defined in the same way as the "PresenceInfo" data type, but with the OpenAPI "nullable: true" property.	
ProblemDetails	3GPP TS 29.571 [11]	Contains a detailed information about an error.	
QosNotifType	3GPP TS 29.514 [17]	Indicates whether the GBR targets for the indicated SDFs are "NOT_GUARANTEED" or "GUARANTEED" again.	
QosResourceType	3GPP TS 29.571 [11]	Indicates whether the resource type is GBR, delay critical GBR, or non-GBR.	
RatingGroup	3GPP TS 29.571 [11]	Identifier of a rating group.	
RatType	3GPP TS 29.571 [11]	The identification of the RAT type.	
RouteToLocation	3GPP TS 29.571 [11]	A traffic routes to applications location.	TSC
ServiceId	3GPP TS 29.571 [11]	Identifier of a service.	
Snsai	3GPP TS 29.571 [11]	Identifies the S-NSSAI.	
SubscribedDefaultQos	3GPP TS 29.571 [11]	Subscribed Default QoS.	
Supi	3GPP TS 29.571 [11]	The identification of the user (i.e. IMSI, NAI).	
SupportedFeatures	3GPP TS 29.571 [11]	Used to negotiate the applicability of the optional features defined in table 5.8-1.	
TraceData	3GPP TS 29.571 [11]		
TimeZone	3GPP TS 29.571 [11]	Contains the user time zone information.	
UInteger	3GPP TS 29.571 [11]	Unsigned Integer.	TimeSensitive Networking
Uri	3GPP TS 29.571 [11]	URI.	
UserLocation	3GPP TS 29.571 [11]	Contains the user location.	
Volume	3GPP TS 29.122 [32]	Unsigned integer identifying a volume in units of bytes.	
VolumeRm	3GPP TS 29.122 [32]	This data type is defined in the same way as the "VolumeRm" data type, but with the OpenAPI "nullable: true" property.	
NOTE 1: "AnGwAddr" data structure is only applicable to the 5GS and EPC/E-UTRAN interworking scenario as defined in Annex B.			
NOTE 2: In order to support a set of MAC addresses with a specific range in the traffic filter, feature MacAddressRange as specified in subclause 5.8 shall be supported.			

*** 2nd Change ***

5.6.2.26 Type UeCampingRep

Table 5.6.2.26-1: Definition of type UeCampingRep

Attribute name	Data type	P	Cardinality	Description	Applicability
accessType	AccessType	O	0..1	The Access Type where the served UE is camping.	
ratType	RatType	O	0..1	The RAT Type where the served UE is camping.	
servNfId	ServingNfIdentity	O	0..1	Contains the serving network function identity.	
servingNetwork	PlmnIdNid	O	0..1	The serving network where the served UE is camping. For an SNPN the NID together with the PLMN ID identifies the SNPN.	
userLocationInfo	UserLocation	O	0..1	The location of the served UE is camping.	
ueTimeZone	TimeZone	O	0..1	The time zone where the served UE is camping.	
netLocAccSupp	NetLocAccessSupport	O	0..1	Indicates the access network does not support the report of the requested access network information. The absence of this attribute indicates that the access network supports access network information reporting.	NetLoc

5

*** 3rd Change ***

5.6.3.x1 Enumeration: NetLocAccessSupport

Table 5.6.3.x1-1: Enumeration NetLocAccessSupport

Enumeration value	Description	Applicability
ANR_NOT_SUPPORTED	Indicates that the access network does not support the report of access network information.	
TZR_NOT_SUPPORTED	Indicates that the access network does not support the report of UE time zone. (NOTE)	
NOTE: The UE time zone is not available in EPC untrusted WLAN.		

10

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*** 4th Change ***

A.2 Npcf_SMPolicyControl API

```

openapi: 3.0.0
info:
  title: Npcf_SMPolicyControl API
  version: 1.1.1.alpha-4
  description: |
    Session Management Policy Control Service
    © 2019, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.
externalDocs:
  description: 3GPP TS 29.512 V16.3.0; 5G System; Session Management Policy Control Service.
  url: 'http://www.3gpp.org/ftp/Specs/archive/29_series/29.512/'
:
  UeCampingRep:
    type: object
    properties:
      accessType:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
      ratType:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/RatType'
      servNfId:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/ServingNfIdentity'
      servingNetwork:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/PlmnIdNid'
      userLocationInfo:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/UserLocation'
      ueTimeZone:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/TimeZone'
      netLocAccSupp:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/NetLocAccessSupport'
  :
  #
  NetLocAccessSupport:
    anyOf:
      - type: string
        enum:
          - ANR_NOT_SUPPORTED
          - TZR_NOT_SUPPORTED
      - type: string
        description: >
          This string provides forward-compatibility with future
          extensions to the enumeration but is not used to encode
          content defined in the present version of this API.
        description: >
          Possible values are
          - ANR_NOT_SUPPORTED: Indicates that the access network does not support the report of
            access network information.
          - TZR_NOT_SUPPORTED: Indicates that the access network does not support the report of UE
            time zone.
  #

```

Fig. 14G

*** 5th Change ***

B.3.2.0 General

When the UE establishes the PDN connection through the EPC network and the SMF+PGW-C receives the Create Session Request message as defined in

- 5 3GPP TS 29.274 [37], the SMF+PGW-C shall behave as defined in subclause 4.2.2.2 with the differences that the SMF+PGW-C shall include (if available) in SmPolicyContextData data structure:

- the IMSI of the user within the "supi" attribute;
- the MSISDN of the user within the "gpsi" attribute;
- 10 - APN within the "dnn" attribute;
- PDU Session Id determined by the SMF+PGW-C within "pduSessionId" attribute for a UE that has an EPS subscription that allows 5GC interworking but does not support 5GC NAS.

NOTE 1: For a PDN connection established via the MME or ePDG, the PDU Session ID value is assigned from a reserved range as specified in Table 5.4.2-1 of TS 29.571 [11];

- 15
- PDN Type within the "pduSessionType" attribute;
 - IMEI-SV within the "pei" attribute;
 - IP-CAN type within the "accessType" attribute;
 - RAT type within the "ratType" attribute;

NOTE 2: See Annex B.3.2.2 for further information.

- 20
- subscribed APN-AMBR within "subsSessAmbr" attribute;
 - subscribed Default EPS bearer QoS within "subsDefQos" attribute;

NOTE 3: Subscribed APN-AMBR and the QCI within the subscribed default EPS bearer QoS are mapped to subscribed Session-AMBR and 5QI as defined in Annex B.3.6.1 respectively.

- user location information within the "userLocationInfo" attribute;
- 25
- NOTE 4: See Annex B.3.2.1 for further information.
- the S-NSSAI determined by the SMF+PGW-C within the "sliceInfo" attribute; and
- the bearer usage required of the default bearer within the "qosFlowUsage" attribute.
- the UE time zone information within "ueTimeZone" attribute, if available;

NOTE x1: The UE time zone is not available in EPC untrusted WLAN.

30

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Fig. 14H

*** 6th Change ***

B.3.6.x1 Request and Report of Access Network information

If the NetLoc feature as defined in subclause 5.8 is supported, the PCF may request the
5 SMF+PGW-C to report the access network information as defined in
subclause 4.2.6.5.4.

If the AN_INFO policy control request trigger is set, upon receiving the
"lastReqRuleData" attribute with the "reqData" attribute with the value(s)
MS_TIME_ZONE and/or USER_LOC_INFO and the "refPccRuleIds" attribute containing
10 the PCC rule identifier(s) corresponding to the PCC rule(s) being installed, modified or
removed:

- If the SMF+PGW-C determines that the access network does not support the access network information
reporting based on the feature support, the SMF+PGW-C shall immediately inform the PCF by including the
15 "netLocAccSupp" attribute set to "ANR_NOT_SUPPORTED" value in the "UeCampingRep" data structure
returned in the "200 OK" response to the policy update notification request.

- Otherwise:

- a) If the "reqData" attribute includes the MS_TIME_ZONE value and the SMF+PGW-C determines that the
access network does not support the report of the UE time zone, the SMF+PGW-C shall immediately
20 inform the PCF by including the "netLocAccSupp" attribute set to "TZR_NOT_SUPPORTED" value in
the "UeCampingRep" data structure returned in the "200 OK" response to the policy update notification
request.

- b) If the "reqData" attribute includes:

- 1) the USER_LOC_INFO value; and/or

- 2) the MS_TIME_ZONE value and the SMF+PGW-C determines the access network supports the report
25 of UE time zone,

the SMF+PGW-C shall apply appropriate procedures to the EPC access network to obtain the requested
access network information and shall behave as specified in subclause 4.2.4.9.

Fig. 14I

***** Start of First Change to 3GPP 29.514 *****

*** 1st Change ***

4.2.2.14 Request of access network information

5 This procedure is used by an AF to request the PCF to report the access network information (i.e., user location and/or user timezone information) at the creation of the "Individual Application Session Context" resource, when the "NetLoc" feature is supported.

The AF shall include in the HTTP POST request message described in

10 subclause 4.2.2.2:

- an entry of the "AfEventSubscription" data type in the "events" attribute with:
 - a) the "event" attribute set to the value "ANI_REPORT"; and
 - b) the "notifMethod" attribute set to the value "ONE_TIME"; and
- 15 - the "reqAni" attribute, with the required access network information, i.e. user location and/or user time zone information).

When the PCF determines that the access network does not support the access network information reporting because the SMF does not support the NetLoc feature, the PCF shall respond to the AF including in the "EventsNotification" data type the "netLocAccSupp" attribute set to "ANR_NOT_SUPPORTED" value. Otherwise, the PCF

20 shall immediately configure the SMF to provide such access information, as specified in 3GPP TS 29.512 [8].

The PCF shall reply to the AF with an HTTP response message as described in subclause 4.2.2.2.

25

30

35

Fig. 15A

*** 2nd Change ***

4.2.3.14 Request of access network information

This procedure is used by an AF to request access network information for an existing "Individual Application Session Context" resource at service information modification
5 when the "NetLoc" feature is supported.

NOTE 1: Subclause 4.2.6.6 describes the AF request of access network information without providing service information.

The AF shall create event subscription information by including in the "AppSessionContextUpdateData" data type the "evSubsc" attribute of
10 "EventsSubscReqData" data type with the corresponding list of events to subscribe to.

The AF shall include in the HTTP PATCH request message described in subclause 4.2.3.2:

- an entry of the "AfEventSubscription" data type in the "events" attribute with:
 - a) the "event" attribute set to the value "ANI_REPORT"; and
 - 15 b) the "notifMethod" attribute set to the value "ONE_TIME"; and
- the "reqAni" attribute, with the required access network information, i.e. user location and/or user time zone information).

When the PCF determines that the access network does not support the access network information reporting because the SMF does not support the NetLoc feature,
20 the PCF shall respond to the AF including in the "EventsNotification" data type the "netLocAccSupp" attribute set to "ANR_NOT_SUPPORTED" value. Otherwise, the PCF shall immediately configure the SMF to provide such access information, as specified in 3GPP TS 29.512 [8].

The PCF shall reply to the AF with an HTTP response message as described in
25 subclause 4.2.3.2.

NOTE 2: The AF does not invoke the Npcf_PolicyAuthorization_Update service operation to remove subscription to access network information report since the "Access Network Information Notification" is the one-time reported event. Once the access network information is reported to the
30 AF the subscription to the access network information report is automatically terminated in the PCF and the related information is removed.

*** 3rd Change ***

4.2.4.6 Request and report of access network information

This procedure is used by an AF to request the PCF to report the access network information (i.e., user location and/or user timezone information) at the deletion of the

- 5 "Individual Application Session Context" resource when the "NetLoc" feature is supported.

This procedure is initiated when:

- the "Individual Application Session Context" is deleted by the AF; or
- 10 - the PCF requests the deletion of the "Individual Application Session Context" from the AF, as described in subclause 4.2.5.3, due to PDU session termination or the termination of all the service data flows of the AF session.

The AF shall include in the HTTP POST request message described in subclause 4.2.4.2:

- an entry of the "AfEventSubscription" data type in the "events" attribute with:
 - 15 a) the "event" attribute set to the value "ANI_REPORT"; and
 - b) the "notifMethod" attribute set to the value "ONE_TIME"; and
- the "reqAni" attribute, with the required access network information, i.e. user location and/or user time zone information).

- 20 When the PCF determines that the access network does not support the access network information reporting because the SMF does not support the NetLoc feature, the PCF shall respond to the AF including in the "EventsNotification" data type the "netLocAccSupp" attribute set to "ANR_NOT_SUPPORTED" value. Otherwise, the PCF shall immediately configure the SMF to provide such access information, as specified in 3GPP TS 29.512 [8].

- 25 When the PCF receives the access network information from the SMF, the PCF shall provide the corresponding access network information to the AF by including the "EventsNotification" data type in the "200 OK" response to the HTTP POST request.

The PCF shall include:

- 30 - in case of 3GPP access, the user location information in the "eutraLocation" or in the "nrLocation" attribute in the "ueLoc" attribute, if available and required;
- in case of untrusted non-3GPP access, the user location information in the "n3gaLocation" attribute in the "ueLoc" attribute, if required, as follows:
 - a) the user local IP address in the "ueIpv4Addr" or "ueIpv6Addr" attribute, if available;
 - b) the UDP source port in the "portNumber" attribute if available; and

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(Continued in Fig. 15D)

Fig. 15C

(Continued from Fig. 15C)

- c) the TCP source port in the "portNumber" attribute if available;
- in case of trusted non-3GPP access, the user location information in the "n3gaLocation" attribute in the "ueLoc" attribute, if required, as follows:
 - 5 a) the user local IP address in the "ueIpv4Addr" or "ueIpv6Addr" attribute, if available;
 - b) the UDP source port in the "portNumber" attribute if available;
 - c) the TCP source port in the "portNumber" attribute if available; and
 - d) the TNAP identifier, that shall consist of:
 - i. the SSID in the "ssid" attribute;
 - 10 ii. the BSSID the "bssid" attribute if available; and
 - iii. the TNAP civic address in the "tnapCivicAddress" attribute if available;
- the serving PLMN network code and country code in the "plmnId" attribute, if user location information is required but not available in any access;
- the UE time zone in the "ueTimeZone" attribute if required and available; and/or
- 15 - the "netLocAccSupp" attribute set to "TZR_NOT_SUPPORTED" value, if UE time zone information is required but not available in current access.

When the PCF receives from the SMF that the access network does not support access network information report, the PCF shall include the "netLocAccSupp" attribute set to "ANR_NOT_SUPPORTED" value in the "EventsNotification" data type in the "200 OK" response to the HTTP POST request.

20 The PCF shall also include an event of the "AfEventNotification" data type in the "evNotifs" attribute with the "event" attribute set to the value "ANI_REPORT".

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Fig. 15D

*** 4th Change ***

4.2.5.11 Reporting access network information

This procedure is used by the PCF to report the access network information (i.e., user location and/or user timezone information) to the AF when the "NetLoc" feature is supported.

When the PCF receives the access network information from the SMF, the PCF shall include the "EventsNotification" data type in the body of the HTTP POST request message sent to the AF as described in subclause 4.2.5.2. The PCF shall include in the "EventsNotification" data type:

- in case of 3GPP access, the user location information in the "utraLocation" or in the "nrLocation" attribute in the "ueLoc" attribute, if available and required;
- in case of untrusted non-3GPP access, the user location information in the "n3gaLocation" attribute in the "ueLoc" attribute, if required, as follows:
 - a) the user local IP address in the "ueIpv4Addr" or "ueIpv6Addr" attribute, if available;
 - b) the UDP source port in the "portNumber" attribute if available; and
 - c) the TCP source port in the "portNumber" attribute if available;
- in case of trusted non-3GPP access, the user location information in the "n3gaLocation" attribute in the "ueLoc" attribute, if required, as follows:
 - a) the user local IP address in the "ueIpv4Addr" or "ueIpv6Addr" attribute, if available;
 - b) the UDP source port in the "portNumber" attribute if available;
 - c) the TCP source port in the "portNumber" attribute if available; and
 - d) the TNAP identifier, that shall consist of:
 - i. the SSID in the "ssid" attribute;
 - ii. the BSSID the "bssid" attribute if available; and
 - iii. the TNAP civic address in the "tnapCivicAddress" attribute if available;
- the serving PLMN network code and country code in the "plmnId" attribute, if user location information is required but not available in any access;
- the UE timezone in the "ueTimeZone" attribute if required and available; and/or
- the "netLocAccSupp" attribute set to "TZR_NOT_SUPPORTED" value, if UE time zone information is required but not available in current access.

When the PCF receives from the SMF that the access network does not support access network information report, the PCF shall include the "netLocAccSupp" attribute set to "ANR_NOT_SUPPORTED" value in the "EventsNotification" data type in the body of the HTTP POST request message sent to the AF as described in subclause 4.2.5.2.

(Continued in Fig. 15F)

Fig. 15E

(Continued from Fig. 15E)

The PCF shall also include an event of the "AfEventNotification" data type in the "evNotifs" attribute with the "event" attribute set to the value "ANI_REPORT".

5 NOTE: The PCF receives the access network information from the SMF if it is previously requested by the AF or at PDU session termination or at the termination of all the service data flows of the AF session.

The PCF shall not invoke the Npcf_PolicyAuthorization_Notify service operation with the "event" attribute set to the value "ANI_REPORT" to report to the AF any subsequently received access network information, unless the AF sends a new request for access network information.

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*** 5th Change ***

4.2.6.6 Request of access network information

This procedure is used by an AF to request the PCF to report the access network information (i.e., user location and/or user timezone information) without providing service information when the "NetLoc" feature is supported.

15 The AF can request access network information without providing service information:

- at initial subscription to events, using the HTTP POST request message as described in subclause 4.2.6.3; and
- at modification of the subscription to events, using the HTTP PUT request message as described in subclause 4.2.6.2.

The AF shall include in the HTTP request message:

20

- an entry of the "AfEventSubscription" data type in the "events" attribute with:
 - a) the "event" attribute set to the value "ANI_REPORT"; and
 - b) the "notifMethod" attribute set to the value "ONE_TIME"; and
- the "reqAni" attribute, with the required access network information, i.e. user location and/or user time zone information).

25 When the PCF determines that the access network does not support the access network information reporting because the SMF does not support the NetLoc feature, the PCF shall respond to the AF including in the "EventsNotification" data type the "netLocAccSupp" attribute set to "ANR_NOT_SUPPORTED" value. Otherwise, the PCF shall immediately configure the SMF to provide such access information, as specified in 3GPP TS 29.512 [8].

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The PCF shall reply to the AF with the HTTP POST response as described in subclause 4.2.6.3 and with the HTTP PUT response as described in subclause 4.2.6.2.

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Fig. 15F

*** 6th Change ***

5.6.1 General

This subclause specifies the application data model supported by the API.

Table 5.6.1-1 specifies the data types defined for the Npcf_PolicyAuthorization service based interface protocol.

Table 5.6.1-1: Npcf_PolicyAuthorization specific Data Types

Data type	Section defined	Description	Applicability
AcceptableServiceInfo	5.6.2.30	Acceptable maximum requested bandwidth.	
AccessNetChargingIdentifier	5.6.2.32	Contains the access network charging identifier.	IMS_SBI
AfEvent	5.6.3.7	Represents an event to notify to the AF.	
AfEventNotification	5.6.2.11	Represents the notification of an event.	
AfEventSubscription	5.6.2.10	Represents the subscription to events.	
AfNotifMethod	5.6.3.8	Represents the notification methods that can be subscribed for an event.	
AfRequestedData	5.6.3.18	Represents the information the AF requested to be exposed.	IMS_SBI
AfRoutingRequirement	5.6.2.13	Describes the routing requirements for the application traffic flows.	InfluenceOnTrafficRouting
AfRoutingRequirementRm	5.6.2.24	This data type is defined in the same way as the "AfRoutingRequirement" data type, but with the OpenAPI "nullable: true" property.	InfluenceOnTrafficRouting
AnGwAddress	5.6.2.20	Carries the control plane address of the access network gateway.	
AppSessionContext	5.6.2.2	Represents an Individual Application Session Context resource.	
AppSessionContextReqData	5.6.2.3	Represents the Individual Application Session Context resource data received in an HTTP POST request message.	
AppSessionContextRespData	5.6.2.4	Represents the Individual Application Session Context resource data produced by the server and returned in an HTTP response message.	
AppSessionContextUpdateData	5.6.2.5	Describes the modifications to an Individual Application Session Context resource.	
ContentVersion	5.6.3.2	Represents the version of a media component.	MediaComponentVersioning
EthFlowDescription	5.6.2.17	Defines a packet filter for an Ethernet flow.	
EventsNotification	5.6.2.9	Describes the notification about the events occurred within an Individual Application Session Context resource.	
EventsSubscReqData	5.6.2.6	Identifies the events the application subscribes to within an Individual Application Session Context resource.	
EventsSubscReqDataRm	5.6.2.6.25	This data type is defined in the same way as the "EventsSubscReqData" data type, but with the OpenAPI "nullable: true" property.	
ExtendedProblemDetails	5.6.2.29	Data type that extends ProblemDetails.	
FlowDescription	5.6.3.2	Defines a packet filter for an IP flow.	

(Continued in Fig. 15H)

(Continued from Fig. 15G)

Flows	5.6.2.21	Identifies the flows related to a media component.	
FlowStatus	5.6.3.12	Describes whether the IP flow(s) are enabled or disabled.	
FlowUsage	5.6.3.14	Describes the flow usage of the flows described by a media subcomponent.	
MediaComponent	5.6.2.7	Contains service information for a media component of an AF session.	
MediaComponentRm	5.6.2.6.26	This data type is defined in the same way as the "MediaComponent" data type, but with the OpenAPI "nullable: true" property.	
MediaComponentResourcesStatus	5.6.3.13	Indicates whether the media component is active or inactive.	
MediaSubComponent	5.6.2.8	Contains the requested bitrate and filters for the set of IP flows identified by their common flow identifier.	
MediaSubComponentRm	5.6.2.27	This data type is defined in the same way as the "MediaSubComponent" data type, but with the OpenAPI "nullable: true" property.	
OutOfCreditInformation	5.6.2.33	Indicates the service data flows without available credit and the corresponding termination action.	IMS_SBI
PcscfRestorationRequestData	5.6.2.36	Indicates P-CSCF restoration.	PCSCF-Restoration-Enhancement
PreemptionControlInformation	5.6.3.19	Pre-emption control information.	MCPTT-Preemption
PreemptionControlInformationRm	5.6.3.21	This data type is defined in the same way as the "PreemptionControlInformation" data type, but with the OpenAPI "nullable: true" property.	MCPTT-Preemption
PrioritySharingIndicator	5.6.3.20	Priority sharing indicator.	PrioritySharing
QosMonitoringInformation	5.6.2.34	QoS monitoring for UL, DL or round trip delay.	QoSMonitoring
QosNotificationControlInfo	5.6.2.15	Indicates whether the QoS targets related to certain media component are not guaranteed or are guaranteed again.	
QosNotifType	5.6.3.9	Indicates type of notification for QoS Notification Control.	
RequiredAccessInfo	5.6.3.15	Indicates the access network information required for an AF session.	NetLoc
ResourcesAllocationInfo	5.6.2.14	Indicates the status of the PCC rule(s) related to certain media component.	
ServAuthInfo	5.6.3.5	Indicates the result of the Policy Authorization service request from the AF.	
ServiceInfoStatus	5.6.3.16	Preliminary or final service information status.	IMS_SBI
ServiceUrn	5.6.3.2	Service URN.	IMS_SBI
SipForkingIndication	5.6.3.17	Describes if several SIP dialogues are related to an "Individual Application Session Context" resource.	IMS_SBI
SpatialValidity	5.6.2.16	Describes the spatial validity of an AF request for influencing traffic routing.	InfluenceOnTrafficRouting
SpatialValidityRm	5.6.2.28	This data type is defined in the same way as the "SpatialValidity" data type, but with the OpenAPI "nullable: true" property.	InfluenceOnTrafficRouting
SponId	5.6.3.2	Contains an identity of a sponsor.	SponsoredConnectivity
SponsoringStatus	5.6.3.6	Represents whether sponsored data connectivity is enabled or disabled/not enabled.	SponsoredConnectivity

(Continued in Fig. 15I)

(Continued from Fig. 15H)

TemporalValidity	5.6.2.22	Indicates the time interval during which the AF request is to be applied.	InfluenceOnTrafficRouting
TerminationCause	5.6.3.10	Indicates the cause for requesting the deletion of the Individual Application Session Context resource.	
TerminationInfo	5.6.2.12	Includes information related to the termination of the Individual Application Session Context resource.	
TosTrafficClass	5.6.3.2	Contains the IPv4 Type-of-Service or the IPv6 Traffic-Class field and the ToS/Traffic Class mask field.	
TosTrafficClassRm	5.6.3.2	This data type is defined in the same way as the "TosTrafficClass" data type, but with the OpenAPI "nullable: true" property.	
TsnQoSContainer	5.6.2.35	TSN QoS parameters.	TimeSensitiveNetworking
UeIdentityInfo	5.6.2.31	Represents 5GS-Level UE Identities.	IMS_SBI

Table 5.6.1-2 specifies data types re-used by the Npcf_PolicyAuthorization service based interface protocol from other specifications, including a reference to their
 5 respective specifications and when needed, a short description of their use within the Npcf_PolicyAuthorization service based interface.

Table 5.6.1-2: Npcf_PolicyAuthorization re-used Data Types

Data type	Reference	Comments	Applicability
AccNetChargingAddress	3GPP TS 29.512 [8]	Indicates the IP address of the network entity within the access network performing charging.	IMS_SBI
AccessType	3GPP TS 29.571 [12]	The identification of the type of access network.	
AccumulatedUsage	3GPP TS 29.122 [15]	Accumulated Usage.	SponsoredConnectivity
AfSigProtocol	3GPP TS 29.512 [8]	Represents the protocol used for signalling between the UE and the AF.	ProvAFsignalFlow
ApplicationChargingId	3GPP TS 29.571 [12]	Application provided charging identifier allowing correlation of charging information.	IMS_SBI
BdtReferenceId	3GPP TS 29.122 [15]	Identifies transfer policies.	
BitRate	3GPP TS 29.571 [12]	Specifies bitrate in kbits per second.	
BitRateRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "BitRate" data type, but with the OpenAPI "nullable: true" property.	
Bytes	3GPP TS 29.571 [12]	String with format "byte".	TimeSensitiveNetworking
ChargingId	3GPP TS 29.571 [12]	Charging identifier allowing correlation of charging information.	IMS_SBI
DateTime	3GPP TS 29.571 [12]	String with format "date-time" as defined in OpenAPI Specification [11].	InfluenceOnTrafficRouting
Dnai	3GPP TS 29.571 [12]	Data network access identifier.	InfluenceOnTrafficRouting
Dnn	3GPP TS 29.571 [12]	Data Network Name.	
FinalUnitAction	3GPP TS 32.291 [22]	Indicates the action to be taken when the user's account cannot cover the service cost.	

(Continued in Fig. 15J)

(Continued from Fig. 15I)

FlowDirection	3GPP TS 29.512 [8]	Flow Direction.	
Gpsi	3GPP TS 29.571 [12]	Identifies the GPSI.	
Ipv4Addr	3GPP TS 29.571 [12]	Identifies an IPv4 address.	
Ipv6Addr	3GPP TS 29.571 [12]	Identifies an IPv6 address.	
MacAddr48	3GPP TS 29.571 [12]	MAC Address.	
NetLocAccessSupport	3GPP TS 29.512 [8]	Indicates the access network support of the report of the requested access network information.	NetLoc
PacketLossRateRm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "PacketLossRate" data type, but with the OpenAPI "nullable: true" property.	CHEM
Pei	3GPP TS 29.571 [12]	Identifies the PEI.	IMS_SBI
PlmnId	3GPP TS 29.571 [12]	PLMN mobile country code and mobile network code.	
PreemptionCapability	3GPP TS 29.571 [12]	Pre-emption capability.	MCPTT-Preemption
PreemptionVulnerability	3GPP TS 29.571 [12]	Pre-emption vulnerability.	MCPTT-Preemption
PreemptionCapabilityRm	3GPP TS 29.571 [12]	It is defined in the same way as the "PreemptionCapability" data type, but with the OpenAPI "nullable: true" property.	MCPTT-Preemption
PreemptionVulnerabilityRm	3GPP TS 29.571 [12]	It is defined in the same way as the "PreemptionVulnerability" data type, but with the OpenAPI "nullable: true" property.	MCPTT-Preemption
PresenceInfo	3GPP TS 29.571 [12]	Represents an area of interest, e.g. a Presence Reporting Area.	InfluenceOnTrafficRouting
RanNasRelCause	3GPP TS 29.512 [8]	Indicates RAN and/or NAS release cause code information.	RAN-NAS-Cause
RatType	3GPP TS 29.571 [12]	RAT Type.	
Snsai	3GPP TS 29.571 [12]	Identifies the S-NSSAI.	
Supi	3GPP TS 29.571 [12]	Identifies the SUPI.	
SupportedFeatures	3GPP TS 29.571 [12]	Used to negotiate the applicability of the optional features defined in table 5.8-1.	
TimeZone	3GPP TS 29.571 [12]	Time Zone.	NetLoc
TsnBridgeInfo	3GPP TS 29.512 [8]	TSN bridge information.	TimeSensitiveNetworking
UInteger	3GPP TS 29.571 [12]	Unsigned Integer, i.e. only value 0 and integers above 0 are permissible. In an OpenAPI Specification [11] schema, the format shall be designated as "UInteger".	
Uint32	3GPP TS 29.571 [12]	Unsigned 32-bit integers, i.e. only value 0 and 32-bit integers above 0 are permissible.	ResourceSharing
Uint32Rm	3GPP TS 29.571 [12]	This data type is defined in the same way as the "Uint32" data type, but with the OpenAPI "nullable: true" property.	ResourceSharing
UpPathChgEvent	3GPP TS 29.512 [8]	Contains the subscription information to be delivered to SMF for the UP path management events.	InfluenceOnTrafficRouting
Uri	3GPP TS 29.571 [12]	String providing an URI.	
UsageThreshold	3GPP TS 29.122 [15]	Usage Thresholds.	SponsoredConnectivity
UsageThresholdRm	3GPP TS 29.122 [15]	This data type is defined in the same way as the "UsageThreshold" data type, but with the OpenAPI "nullable: true" property.	SponsoredConnectivity
UserLocation	3GPP TS 29.571 [12]	User Location.	NetLoc

Fig. 15J

*** 7th Change ***

5.6.2.9 Type EventsNotification**Table 5.6.2.9-1: Definition of type EventsNotification**

Attribute name	Data type	P	Cardinality	Description	Applicability
accessType	AccessType	C	0..1	Includes the access type. It shall be present when the notified event is "ACCESS_TYPE_CHG".	
anChargAddr	AccNetChargingAddress	O	0..1	Includes the access network charging address. It shall be present if available when the notified event is "CHARGING_CORRELATION".	IMS_SBI
anChargIds	array(AccessNetChargingIdentifier)	C	1..N	Includes the access network charging identifier(s). It shall be present when the notified event is "CHARGING_CORRELATION".	IMS_SBI
anGwAddr	AnGwAddress	O	0..1	Access network Gateway Address. It shall be present, if applicable, when the notified event is "ACCESS_TYPE_CHG".	
evSubsUri	Uri	M	1	The Events Subscription URI. Identifies the Events Subscription sub-resource that triggered the notification.	
evNotifs	array(AfEventNotification)	M	1..N	Notifications about individual events.	
failedResourceAllocationReports	array(ResourcesAllocationInfo)	C	1..N	Indicates the status of the PCC rule(s) related to certain failed media components. It shall be included when the event trigger is "FAILED_RESOURCES_ALLOCATION".	
netLocAccSupport	NetLocAccessSupport	O	0..1	Indicates the access network does not support the report of the requested access network information. The absence of this attribute indicates that the access network supports access network information reporting.	NetLoc
outOfCredReports	array(OutOfCreditInformation)	C	1..N	Out of credit information per service data flow. It shall be present when the notified event is "OUT_OF_CREDIT".	IMS_SBI
plmnId	PlmnId	C	0..1	PLMN Identifier. It shall be present when the notified event is "PLMN_CHG" or, if location information is required but is not available when the notified event is "ANI_REPORT". It shall be present if available when the notified event is "RAN_NAS_CAUSE".	
qncReports	array(QosNotificationControlInfo)	C	1..N	QoS notification control information. It shall be present when the notified event is "QOS_NOTIF".	
ranNasRelCauses	array(RanNasRelCause)	C	1..N	RAN-NAS release cause. It shall be present if available when the notified event is "RAN_NAS_CAUSE".	RAN-NAS-Cause
ratType	RatType	O	0..1	RAT type. It shall be present, if applicable, when the notified event is "ACCESS_TYPE_CHG".	

(Continued in Fig. 15L)

(Continued from Fig. 15K)

ueLoc	UserLocation	O	0..1	E-UTRA, NR, or non-3GPP trusted and untrusted access user location information. "n3gppTai" and "n3lwfld" attributes within the "N3gaLocation" data type shall not be supplied. It shall be present if required and available when the notified event is "ANI_REPORT". It shall be present if available when the notified event is "RAN_NAS_CAUSE".	NetLoc, RAN-NAS-Cause
ueTimeZone	TimeZone	O	0..1	UE time zone. It shall be present if required and available when the notified event is "ANI_REPORT". It shall be present if available when the notified event is "RAN_NAS_CAUSE".	NetLoc, RAN-NAS-Cause
usgRep	AccumulatedUsage	C	0..1	Indicates the measured volume and/or time for sponsored data connectivity. It shall be present when the notified event is "USAGE_REPORT".	SponsoredConnectivity
tsnBridgeInfo	TsnBridgeInfo	O	0..1	Reports the TSN bridge information.	TimeSensitiveNetworking
tsnPortManContD stt	Bytes	O	0..1	Transports TSN port management information encoded as specified in subclause 9.11.4.27 of 3GPP TS 24.501 [49] starting with octet 2.	TimeSensitiveNetworking
tsnPortManContN wtt	Bytes	O	0..1	Transports TSN port management information encoded as specified in subclause 9.11.4.27 of 3GPP TS 24.501 [49] starting with octet 2.	TimeSensitiveNetworking

*** 8th Change ***

5

A.2 Npcf_PolicyAuthorization API

```

openapi: 3.0.0
info:
  title: Npcf_PolicyAuthorization Service API
  version: 1.1.0.alpha-4
  description: |
    PCF Policy Authorization Service.
    © 2019, 3GPP Organizational Partners (ARIB, ATIS, CCSA, ETSI, TSDSI, TTA, TTC).
    All rights reserved.
externalDocs:
  description: 3GPP TS 29.514 V16.3.0; 5G System; Policy Authorization Service; Stage 3.
  url: 'http://www.3gpp.org/ftp/Specs/archive/29_series/29.514/'
#
:
  EventsNotification:
    description: describes the notification of a matched event
    type: object
    required:
      - evSubsUri
      - evNotifs
    properties:
      accessType:
        $ref: 'TS29571_CommonData.yaml#/components/schemas/AccessType'
      anChargAddr:
        $ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/AccNetChargingAddress'

```

(Continued in Fig. 15M)

Fig. 15L

(Continued from Fig. 15L)

5 anChargId:
 type: array
 items:
 \$ref: '#/components/schemas/AccessNetChargingIdentifier'
 minItems: 1
 anGwAddr:
 \$ref: '#/components/schemas/AnGwAddress'
 evSubsUri:
 \$ref: 'TS29571_CommonData.yaml#/components/schemas/Uri'
 evNotifs:
 type: array
 items:
 \$ref: '#/components/schemas/AnEventNotification'
 minItems: 1
 failedResourceAllocReports:
 type: array
 items:
 \$ref: '#/components/schemas/ResourcesAllocationInfo'
 minItems: 1
 netLocAccSupp:
 \$ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/NetLocAccessSupport'
 outOfCredReports:
 type: array
 items:
 \$ref: '#/components/schemas/OutOfCreditInformation'
 minItems: 1
 plmnId:
 \$ref: 'TS29571_CommonData.yaml#/components/schemas/PlmnId'
 qncReports:
 type: array
 items:
 \$ref: '#/components/schemas/QosNotificationControlInfo'
 minItems: 1
 ranNasRelCauses:
 type: array
 items:
 \$ref: 'TS29512_Npcf_SMPolicyControl.yaml#/components/schemas/RanNasRelCause'
 minItems: 1
 description: Contains the RAN and/or NAS release cause.
 ratType:
 \$ref: 'TS29571_CommonData.yaml#/components/schemas/RatType'
 ueLoc:
 \$ref: 'TS29571_CommonData.yaml#/components/schemas/UserLocation'
 ueTimeZone:
 \$ref: 'TS29571_CommonData.yaml#/components/schemas/TimeZone'
 usgRep:
 \$ref: 'TS29122_CommonData.yaml#/components/schemas/AccumulatedUsage'
 :
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*** End of Changes ***

55

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Fig. 15M

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PARTIAL SUPPORT OF ACCESS NETWORK INFORMATION

RELATED APPLICATIONS

This application claims the benefit of U.S. provisional patent application Ser. No. 62/977,635, filed Feb. 17, 2020, the disclosure of which is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates to provision of access network information to a requester in a network.

BACKGROUND

Signalling Flows for IMS, Request of NetLoc Information by the P-CSCF

Third Generation Partnership Project (3GPP) has developed NetLoc, which is a feature to make the Network Provided Location Information (NPLI) related to the access network that the user equipment (UE) is camped on available to the internet protocol (IP) multimedia system (IMS) nodes whenever the IMS operator needs to record this information (e.g., to fulfil legal obligations such as a regulation requiring logging call information together with UE Location), for charging purposes or because of other application specific requirements.

Clause 1 describes an example of an application request of network provided location information (there are other IMS/SDP (session description protocol) interactions where the P-CSCF (proxy-call session control function) requests NPLI). Clause 2 describes how the policy control function (PCF) then requests NPLI to the session management function (SMF) (or SMF+PGW (Packet data network (PDN) GateWay). Clauses 2 and 3 describe how the PCF gets the NPLI from the access network. Relevant text for these procedures is set forth below.

Clause 1: Provisioning of Service Information at Originating P-CSCF and PCF

FIG. 1 illustrates conventional policy and charging control (PCC) Procedures for IMS Session Establishment at an originating P-CSCF and a PCF. In FIG. 1, the P-CSCF derives the provisioning of service information to the PCF from the SDP offer/answer exchange. FIG. 1 illustrates the following steps:

- Step 100. The P-CSCF receives the SDP parameters defined by the originator within an SDP offer in session initiation protocol (SIP) signalling.
- Step 102. The P-CSCF identifies and defines the (downlink) connection information needed (IP address of the downlink IP flow(s), port numbers to be used, etc.).
- Step 104. The P-CSCF forwards the SDP offer in SIP signalling.
- Step 106. The P-CSCF gets the negotiated SDP parameters from the terminating side through an SIP signalling interaction within an SDP answer.
- Step 108. The P-CSCF identifies the (uplink) connection information needed (IP address of the uplink media IP flow(s), port numbers to be used, etc.).
- Step 110. The P-CSCF invokes or requests the Npcf_PolicyAuthorization_Create service operation to forward the derived session information to the PCF by sending an HTTP POST request to the “Application Sessions” resource.

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Step 112. (Rx case) The P-CSCF provides session information to the PCF by sending a Diameter authentication and authorization request (AAR) for a new Rx Diameter session.

In these steps (110, 112), the P-CSCF may also request the report of network provided location information (NPLI).

Step 114. The PCF stores application session information and performs session binding and/or identifies a protocol data unit (PDU) session. For an N5 interface, the PCF creates an “Individual Application Session Context” resource to store the received application session information.

Step 116. The PCF replies to the P-CSCF with an HTTP “201 Created” (e.g., Npcf_PolicyAuthorization_Create) response and includes the uniform resource identifier (URI) of the “Individual Application Session Context” resource in the Location header field.

Step 118. (Rx case) The PCF sends a Diameter authentication, authorization, and accounting (AAA) to the P-CSCF.

Step 120. Upon reception of the acknowledgement from the PCF, the SDP parameters are passed to the UE in SIP signalling.

Step 122. The PCF executes interactions according to FIG. 2. This step implies provisioning of PCC rules and is executed in parallel with steps 124 and 126 (steps 128 and 130 for Rx case). The provisioning of PCC rules, if requested by the P-CSCF in step 110 (step 112 for Rx case), shall include the request of reporting network provided location information (NPLI).

Step 124. If the P-CSCF requested access network information in step 110, the PCF invokes the Npcf_PolicyAuthorization_Notify service operation to forward the access network information received in step 122 in an HTTP POST request sent to the Notification URI received in step 110.

Step 126. If step 124 occurs, the P-CSCF acknowledges the receipt of the notification request with an HTTP “204 No Content” (e.g., Npcf_PolicyAuthorization_Notify) response to the PCF.

Step 128. (Rx case) If the P-CSCF requested access network information in step 112, the PCF forwards the access network information received in step 122 in a Diameter re-authorization request (RAR).

Step 130. (Rx case) If step 128 occurs, the P-CSCF acknowledges the receipt of the Diameter RAR through a Diameter Re-authorization Acknowledgment (RAA).

Step 132. If step 124 occurs (step 128 for Rx case), the P-CSCF forwards the access network information as the NPLI when a suitable SIP message is received.

Clause 2: Provisioning of PCC Rules in the SMF (or SMF+PGW)

FIG. 2 illustrates conventional interactions between a PCF and a SMF for PCF-initiated session management (SM) Policy Association Modification procedures. This procedure is performed when the PCF decides to modify the PDU session (updating PCC rules) due to the P-CSCF provisioning of service information.

Step 200. The PCF receives a P-CSCF request to provision service information, as described in FIG. 1, step 110 (step 112 for Rx case), that triggers the re-evaluation of the PCC rule information to install in the SMF (or SMF+PGW).

Step 202. The PCF binds the P-CSCF request with an SM Policy Context (a PDU Session), and determines that a network initiated PDU session modification procedure

is required. New PCC rules, requesting the report of access network information (NPLI), are installed in the SMF.

Step 204. The PCF invokes the Npcf_SMPolicyControl_UpdateNotify service operation by sending the HTTP POST request with “{Notification URI}/update” as the resource URI to the SMF that has previously subscribed. The request operation provides the PDU session identification (ID) and the updated PCC rules, as described in subclause 4.2.3 of 3GPP TS 29.512.

Step 206. The SMF sends an HTTP “200 OK” (e.g., an Npcf_SMPolicyControl_UpdateNotify Response) to the PCF to acknowledge the installation of the PCC rules. The SMF initiates the PDU session modification procedures towards the access and mobility management function (AMF) (or the S-GW/ePDG in case the UE is connected to a 4G Access network). As part of this PDU session modification request, the SMF requests NPLI.

Step 208. When the SMF receives the available NPLI (UE Location and/or UE Time Zone), the SMF reports it to the PCF invoking the Npcf_SMPolicyControl_Update service operation by sending the HTTP POST request with request URI for the Individual SM Policy Context resource URI and an HTTP body including the UE Location and/or UE Time Zone as described in subclause 4.2.4.9 of 3GPP TS 29.512.

Step 210. The PCF sends to the SMF an HTTP “200 OK” (e.g., an Npcf_SMPolicyControl_UpdateResponse) to acknowledge the report and continues with step 124 of FIG. 1.

Clause 3: Access Network Information Reporting in 5G Network

To support charging data collection and to fulfill regulatory requirements (e.g., to provide NPLI as defined in TS 23.228 [15]) related with the set-up, modification, and release of IMS Voice calls or with SMS transfer the following applies for the cases where the UE is accessing via a 5G Network:

When an AMF forwards uplink (UL) non-access stratum (NAS) or N2 signalling to a peer network function (NF) (e.g., to an SMF or to a short message service function (SMSF)) or during the UP connection activation of a PDU Session, the AMF provides any User Location Information it has received from the 5G-access network (AN) as well as the Access Type (3GPP-Non 3GPP) of the AN over which it has received the UL NAS or N2 signalling. The AMF also provides the corresponding UE Time Zone. In addition, to fulfill regulatory requirements (e.g., providing NPLI, as defined in TS 23.228 [15]) when the access is non-3GPP, the AMF may also provide the last known 3GPP access User Location Information with its age, if the UE is still attached to the same AMF for 3GPP access (i.e., valid User Location Information).

The User Location Information, the access type and the UE Time Zone may be further provided by the SMF to the PCF if the PCF has requested so (see clause 1 and clause 2 above). The PCF may get this information from the SMF to provide NPLI to applications (such as IP multimedia subsystem (IMS)) that have requested it.

SUMMARY

Embodiments of a method performed by a Session Management Function (SMF) for providing Access Network (AN) information are disclosed. The method comprises

receiving a request for AN information from a Policy Control Function (PCF). The method also comprises determining whether an AN does not support reporting of the requested AN information. The method also comprises, upon determining that the AN does not support reporting of the requested AN information, sending, to the PCF, a notification that the AN does not support reporting of the requested AN information. The method also comprises, upon determining that the AN does support reporting of the requested AN information, acquiring the requested AN information and sending the requested AN information to the PCF.

In one embodiment, sending the notification that the AN does not support the reporting of the requested AN information comprises sending a UeCampingRep data structure. In one embodiment, sending the notification that the AN does not support the reporting of the requested AN information comprises sending a UeCampingRep data structure having a netLocAccSupp attribute set to an ANR_NOT_SUPPORTED value. In one embodiment, the requested AN information comprises User Equipment (UE) location information or timezone information, and sending the notification that the AN does not support the reporting of the requested AN information comprises sending a UeCampingRep data structure having a netLocAccSupp attribute set to a TZR_NOT_SUPPORTED value.

Embodiments of a method performed by a PCF for providing AN information are also disclosed. The method comprises receiving a request for AN information from an Application Function (AF). The method also comprises forwarding the request to an SMF. The method also comprises receiving, from the SMF, a response to the request. The method also comprises forwarding the response to the AF.

In one embodiment, the response from the SMF comprises an indication that an AN does not support reporting of the requested AN information. In one embodiment, the response from the SMF comprises a UeCampingRep data structure. In one embodiment, the UeCampingRep data structure comprises a netLocAccSupp attribute set to an ANR_NOT_SUPPORTED value. In one embodiment, the requested AN information comprises UE location information or timezone information and the UeCampingRep data structure comprises a netLocAccSupp attribute set to a TZR_NOT_SUPPORTED value.

Embodiments of a method performed by an AF for requesting AN information are disclosed. The method comprises sending a request for AN information to a PCF. The method also comprises receiving, from the PCF, a response to the request for AN information. The response includes the requested AN information or includes an indication that an AN does not support reporting of the requested AN information.

In one embodiment, the response from the PCF comprises a UeCampingRep data structure. In one embodiment, the UeCampingRep data structure comprises a netLocAccSupp attribute set to an ANR_NOT_SUPPORTED value. In one embodiment, the requested AN information comprises UE location information or timezone information and the UeCampingRep data structure comprises a netLocAccSupp attribute set to a TZR_NOT_SUPPORTED value.

Embodiments of an SMF are also disclosed. The SMF is adapted to receive a request for AN information from a PCF. The SMF is also adapted to determine whether an AN does not support reporting of the requested AN information. The SMF is adapted to, upon determining that the AN does not support reporting of the requested AN information, send, to

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the PCF, a notification that the AN does not support reporting of the requested AN information. The SMF is also adapted to, upon determining that the AN does support reporting of the requested AN information, acquire the requested AN information, and send the requested AN information to the PCF. In one embodiment, the SMF is further adapted to perform any of the embodiments of the method performed by an SMF for providing AN information.

Embodiments of a PCF are also disclosed. The PCF is adapted to receive, from an AF, a request for AN information. The PCF is also adapted to forward the request to an SMF. The PCF is also adapted to receive, from the SMF, a response to the request. The PCF is also adapted to forward the response to the AF. In one embodiment, the PCF is further adapted to perform any of the embodiments of the method performed by a PCF for providing AN information.

Embodiment of an AF are also disclosed. The AF is adapted to send, to a PCF, a request for AN information. The AF is also adapted to receive, from the PCF, a response to the request for AN information. The response includes the requested AN information or includes an indication that an AN does not support reporting of the requested AN information. In one embodiment, the AF is further adapted to perform any of the embodiments of the method performed by an AF for requesting AN information.

Embodiments of a network node for implementing an SMF are also disclosed. The network node comprises processing circuitry. The processing circuitry is configured to cause the network node to receive, from a PCF, a request for AN information. The processing circuitry is also configured to cause the network node to determine whether an AN does not support reporting of the requested AN information. The processing circuitry is also configured to cause the network node to, upon determining that the AN does not support reporting of the requested AN information, send, to the PCF, a notification that the AN does not support reporting of the requested AN information. The processing circuitry is also configured to cause the network node to, upon determining that the AN does support reporting of the requested AN information, acquire the requested AN information, and send the requested AN information to the PCF.

In one embodiment, the processing circuitry configured to send the notification that the AN does not support the reporting of the requested AN information is further configured to send a UeCampingRep data structure. In one embodiment, the processing circuitry configured to send the notification that the AN does not support the reporting of the requested AN information is further configured to send a UeCampingRep data structure having a netLocAccSupp attribute set to an ANR_NOT_SUPPORTED value. In one embodiment, the requested AN information comprises UE location information or timezone information, and the processing circuitry configured to send the notification that the AN does not support the reporting of the requested AN information is further configured to send a UeCampingRep data structure having a netLocAccSupp attribute set to a TZR_NOT_SUPPORTED value.

Embodiments of a network node for implementing a PCF are also disclosed. The PCF comprises processing circuitry. The processing circuitry is configured to cause the network node to receive, from an AF, a request for AN information. The processing circuitry is also configured to cause the network node to forward the request to an SMF. The processing circuitry is also configured to cause the network node to receive, from the SMF, a response to the request.

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The processing circuitry is also configured to cause the network node to forward the response to the AF.

In one embodiment, the response from the SMF comprises an indication that an AN does not support reporting the requested AN information. In one embodiment, the response from the SMF comprises a UeCampingRep data structure. In one embodiment, the UeCampingRep data structure comprises a netLocAccSupp attribute set to an ANR_NOT_SUPPORTED value. In one embodiment, the requested AN information comprises UE location information or timezone information and the UeCampingRep data structure comprises a netLocAccSupp attribute set to a TZR_NOT_SUPPORTED value.

Embodiments of a network node for implementing an AF are also disclosed. The AF comprises processing circuitry. The processing circuitry is configured to cause the network node to send, to a PCF, a request for AN information. The processing circuitry is also configured to cause the network node to receive, from the PCF, a response to the request for AN information. The response includes the requested AN information or includes an indication that an AN does not support reporting the requested AN information.

In one embodiment, the response from the PCF comprises a UeCampingRep data structure. In one embodiment, the UeCampingRep data structure comprises a netLocAccSupp attribute set to an ANR_NOT_SUPPORTED value. In one embodiment, the requested AN information comprises UE location information or timezone information and the UeCampingRep data structure comprises a netLocAccSupp attribute set to a TZR_NOT_SUPPORTED value.

Embodiments of a method performed in a cellular communication system are also disclosed. The method comprises, at an SMF for providing AN information, receiving, from a PCF, a request for AN information. The method also comprises determining whether an AN does not support reporting of the requested AN information. The method also comprises, upon determining that the AN does not support reporting of the requested AN information, sending, to the PCF, a response comprising a notification that the AN does not support reporting of the requested AN information. The method also comprises, upon determining that the AN does support reporting of the requested AN information, acquiring the requested AN information, and sending the requested AN information to the PCF. The method also comprises, at the PCF, receiving, from an AF, the request for AN information. The method also comprises forwarding the request to the SMF. The method also comprises receiving, from the SMF, the response to the request. The method also comprises forwarding the response to the AF. The method also comprises, at the AF, sending, to the PCF, the request for AN information. The method also comprises receiving, from the PCF, the response to the request for AN information. The response includes the requested AN information or includes an indication that an AN does not support reporting the requested AN information.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawing figures incorporated in and forming a part of this specification illustrate several aspects of the disclosure, and together with the description serve to explain the principles of the disclosure.

FIG. 1 illustrates conventional policy and charging control (PCC) Procedures for IMS Session Establishment at an originating P-CSCF and a PCF;

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FIG. 2 illustrates conventional interactions between a PCF and a SMF for PCF-initiated session management (SM) Policy Association Modification procedures;

FIG. 3 illustrates one example of a cellular communication network according to some embodiments of the present disclosure;

FIG. 4 illustrates a wireless communication system represented as a 5G network architecture composed of core Network Functions (NFs);

FIG. 5 illustrates a 5G network architecture using service-based interfaces between the NFs in the control plane, instead of the point-to-point reference points/interfaces used in the 5G network architecture of FIG. 4;

FIG. 6 illustrates an exemplary method for partial access of network information according to some embodiments of the present disclosure;

FIG. 7 illustrates an exemplary method for partial access of network information according to some embodiments of the present disclosure;

FIG. 8 is a schematic block diagram of a network node, and particularly a network node according to some embodiments of the present disclosure;

FIG. 9 is a schematic block diagram that illustrates a virtualized embodiment of a network node of FIG. 8 according to some embodiments of the present disclosure;

FIG. 10 is a schematic block diagram of the network node of FIG. 8 according to some other embodiments of the present disclosure;

FIG. 11 is a schematic block diagram of a UE according to some embodiments of the present disclosure;

FIG. 12 is schematic block diagram of the UE of FIG. 11 according to some embodiments of the present disclosure;

FIGS. 13A-13H illustrate changes to 3GPP TS 29.512 V15.6.0 according to some embodiments of the present disclosure;

FIGS. 14A-14I illustrate changes to 3GPP TS 29.512 V16.3.0 according to some embodiments of the present disclosure; and

FIGS. 15A-15M illustrate changes to 3GPP TS 29.514 V16.3.0 according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

The embodiments set forth below represent information to enable those skilled in the art to practice the embodiments and illustrate the best mode of practicing the embodiments. Upon reading the following description in light of the accompanying drawing figures, those skilled in the art will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. It should be understood that these concepts and applications fall within the scope of the disclosure.

Radio Node: As used herein, a “radio node” is either a radio access node or a wireless communication device.

Radio Access Node: As used herein, a “radio access node” or “radio network node” or “radio access network node” is any node in a Radio Access Network (RAN) of a cellular communications network that operates to wirelessly transmit and/or receive signals. Some examples of a radio access node include, but are not limited to, a base station (e.g., a New Radio (NR) base station (gNB) in a Third Generation Partnership Project (3GPP) Fifth Generation (5G) NR network or an enhanced or evolved Node B (eNB) in a 3GPP Long Term Evolution (LTE) network), a high-power or macro base station, a low-power base station (e.g., a micro base station, a pico base station, a home eNB, or the like),

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a relay node, a network node that implements part of the functionality of a base station (e.g., a network node that implements a gNB Central Unit (gNB-CU) or a network node that implements a gNB Distributed Unit (gNB-DU)) or a network node that implements part of the functionality of some other type of radio access node.

Core Network Node: As used herein, a “core network node” is any type of node in a core network or any node that implements a core network function. Some examples of a core network node include, e.g., a Mobility Management Entity (MME), a Packet Data Network Gateway (PGW), a Service Capability Exposure Function (SCEF), a Home Subscriber Server (HSS), or the like. Some other examples of a core network node include a node implementing a Access and Mobility Management Function (AMF), a UPF, a Session Management Function (SMF), an Authentication Server Function (AUSF), a Network Slice Selection Function (NSSF), a Network Exposure Function (NEF), a Network Function (NF) Repository Function (NRF), a Policy Control Function (PCF), a Unified Data Management (UDM), or the like.

Communication Device: As used herein, a “communication device” is any type of device that has access to an access network. Some examples of a communication device include, but are not limited to: mobile phone, smart phone, sensor device, meter, vehicle, household appliance, medical appliance, media player, camera, or any type of consumer electronic, for instance, but not limited to, a television, radio, lighting arrangement, tablet computer, laptop, or Personal Computer (PC). The communication device may be a portable, hand-held, computer-comprised, or vehicle-mounted mobile device, enabled to communicate voice and/or data via a wireless or wireline connection.

Wireless Communication Device: One type of communication device is a wireless communication device, which may be any type of wireless device that has access to (i.e., is served by) a wireless network (e.g., a cellular network). Some examples of a wireless communication device include, but are not limited to: a User Equipment (UE) in a 3GPP network, a Machine Type Communication (MTC) device, and an Internet of Things (IoT) device. Such wireless communication devices may be, or may be integrated into, a mobile phone, smart phone, sensor device, meter, vehicle, household appliance, medical appliance, media player, camera, or any type of consumer electronic, for instance, but not limited to, a television, radio, lighting arrangement, tablet computer, laptop, or PC. The wireless communication device may be a portable, hand-held, computer-comprised, or vehicle-mounted mobile device, enabled to communicate voice and/or data via a wireless connection.

Network Node: As used herein, a “network node” is any node that is either part of the radio access network or the core network of a cellular communications network/system.

Note that the description given herein focuses on a 3GPP cellular communications system and, as such, 3GPP terminology or terminology similar to 3GPP terminology is oftentimes used. However, the concepts disclosed herein are not limited to a 3GPP system.

Note that, in the description herein, reference may be made to the term “cell”; however, particularly with respect to 5G NR concepts, beams may be used instead of cells and, as such, it is important to note that the concepts described herein are equally applicable to both cells and beams.

There currently exist certain challenge(s) when an application function (AF) requests network information. According to current procedures, whenever the AF needs network

provided location information (NPLI) for specific reasons, the AF subscribes to be informed about NPLI: either location information or time zone, or both, depending on the purpose. Additionally, the AF may subscribe to receive information about radio access type (RAT) type/Access Type changes in the policy control function (PCF), if this information is relevant for the AF to request this specific information.

As the functionality is defined today, it is possible that not all the access information is available in the access network. That is:

For Untrusted wireless local area networks (WLANs), the time zone is not available

Any access network could not provide the information for privacy reasons, network policies, impossibility of derivation, etc. If the information refers to the location information, according to current procedures, the serving public land mobile network (PLMN) network code and country code will be provided instead. However, nothing is provided when it refers to the time zone.

When the NetLoc feature is not supported by the session management function (SMF) or the PCF, the AF receives information that the Access Report is not supported.

With this state of the art the following limitations exist:

For access networks where the time zone is not available, the AF will not be notified when the time zone changes or that the access network does not support the reporting. Thus, it can make wrong assumptions and may consider that the time zone has not changed even if it changed.

For access networks where the time zone is not available, it is not specified what the SMF will do when the time zone is not provided by the access network and it was requested by the AF. It means unexpected behavior will occur in the network.

Even if the AF subscribes to access type changes simultaneously with the request of access information (or in advance), the AF will not know if the non-3GPP network is trusted or untrusted or if the network is private. The AF could derive the kind of access network if the AF got the location (whose encoding depends on the kind of 3GPP/non-3GPP trusted/untrusted) but it will mean extra logic in the AF and the mandatory subscription to location information.

The SMF/PCF informs about the lack of support of NetLoc functionality, but the AF will not be informed if the access network can provide the information and if it can be provided only partially.

Certain aspects of the present disclosure and their embodiments may provide solutions to the aforementioned or other challenges. The present disclosure proposes a solution with the following functionality: when the Access Network does not support part of the information required by the AF (e.g., time zone), the SMF will indicate to the PCF that this specific information is not supported and the PCF will indicate so to the AF. As such, the present disclosure provides the following:

A mechanism to allow the AF to get clear information about the availability of all or partial requested access information.

When the SMF cannot get part of the requested information (time zone), the SMF will notify the PCF that this information is not provided because the network does not support it.

When the PCF gets information that part of the requested information (time zone) is not supported, the PCF will inform the AF about this lack of support

There are, proposed herein, various embodiments which address one or more of the issues disclosed herein. Certain embodiments may provide one or more of the following technical advantage(s):

The AF gets clear information on whether all or partial requested information was available or not and the reason why it was not provided.

Whenever the AF subscribes to time zone reporting, the AF will get information of the time zone or, if not supported by the access network or not available, about the lack of support.

The AF will not have to assume that the lack of reporting means that the requested information is not supported, i.e., the SMF will be able to report information about the lack of support of partial information when the AF subscribes to receive only time zone information and the network does not support it.

The AF will not have to interpret that the absence of time zone in the access network reporting means that it was not provided because it was not supported by the access network.

The AF will not have to subscribe to access type changes and to always require the location information in combination with the time zone (even if the location was not needed) to derive that the absence of time zone means that the access network does not support it.

Before addressing particular embodiments of the present disclosure, a general overview of a cellular communications system is provided with reference to FIG. 3. In this regard, FIG. 3 illustrates one example of a cellular communications system 300 in which embodiments of the present disclosure may be implemented. In the embodiments described herein, the cellular communications system 300 is a 5G system (5GS) including a new radio (NR) radio access node (RAN) (NR RAN) or long term evolution (LTE) RAN (i.e., E-UTRA RAN) or an Evolved Packet System (EPS) including an LTE RAN. In this example, the RAN includes base stations 302-1 and 302-2, which in LTE are referred to as enhanced node Bs (eNBs) (when connected to an evolved packet core (EPC)) and in 5G NR are referred to as gNBs (e.g., LTE RAN nodes connected to 5GC, which are referred to as gn-eNBs), controlling corresponding (macro) cells 304-1 and 304-2. The base stations 302-1 and 302-2 are generally referred to herein collectively as base stations 302 and individually as base station 302. Likewise, the (macro) cells 304-1 and 304-2 are generally referred to herein collectively as (macro) cells 304 and individually as (macro) cell 304. The RAN may also include a number of low power nodes 306-1 through 306-4 controlling corresponding small cells 308-1 through 308-4. The low power nodes 306-1 through 306-4 can be small base stations (such as pico or femto base stations) or Remote Radio Heads (RRHs), or the like. Notably, while not illustrated, one or more of the small cells 308-1 through 308-4 may alternatively be provided by the base stations 302. The low power nodes 306-1 through 306-4 are generally referred to herein collectively as low power nodes 306 and individually as low power node 306. Likewise, the small cells 308-1 through 308-4 are generally referred to herein collectively as small cells 308 and individually as small cell 308. The cellular communications system 300 also includes a core network 310, which in the 5GS is referred to as the 5G core (5GC). The base stations 302 (and optionally the low power nodes 306) are connected to the core network 310.

The base stations 302 and the low power nodes 306 provide service to wireless communication devices 312-1 through 312-5 in the corresponding cells 304 and 308. The

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wireless communication devices **312-1** through **312-5** are generally referred to herein collectively as wireless communication devices **312** and individually as wireless communication device **312**. In the following description, the wireless communication devices **312** are oftentimes UEs, but the present disclosure is not limited thereto.

FIG. **4** illustrates a wireless communication system represented as a 5G network architecture composed of core Network Functions (NFs), where interaction between any two NFs is represented by a point-to-point reference point/interface. FIG. **4** can be viewed as one particular implementation of the core network **310** of the system **300** of FIG. **3**.

Seen from the access side, the 5G network architecture shown in FIG. **4** comprises a plurality of UEs **400** connected to either a RAN or an Access Network (AN) **402** as well as an AMF **404**. Typically, the (R)AN **402** comprises base stations, e.g., such as evolved Node Bs (eNBs) or NR base stations (gNBs) or similar. Seen from the core network side, the 5G core NFs shown in FIG. **4** include a Network Slice Selection Function (NSSF) **406**, an Authentication Server Function (AUSF) **408**, a Unified Data Management (UDM) **410**, the AMF **404**, a Session Management Function (SMF) **412**, a Policy Control Function (PCF) **414**, and an Application Function (AF) **416**.

Reference point representations of the 5G network architecture are used to develop detailed call flows in the normative standardization. The N1 reference point is defined to carry signaling between the UE **400** and AMF **404**. The reference points for connecting between the AN **402** and AMF **404** and between the AN **402** and user plane function (UPF) **418** are defined as N2 and N3, respectively. There is a reference point, N11, between the AMF **404** and SMF **412**, which implies that the SMF **412** is at least partly controlled by the AMF **404**. N4 is used by the SMF **412** and UPF **418** so that the UPF **418** can be set using the control signal generated by the SMF **412**, and the UPF **418** can report its state to the SMF **412**. N9 is the reference point for the connection between different UPFs **418**, and N14 is the reference point connecting between different AMFs **404**, respectively. N15 and N7 are defined since the PCF **414** applies policy to the AMF **404** and SMF **412**, respectively. N12 is required for the AMF **404** to perform authentication of the UE **400**. N8 and N10 are defined because the subscription data of the UE **400** is required for the AMF **404** and SMF **412**. N6 is defined between the UPF **418** and a data network (DN) **420**.

The 5G core network aims at separating the user plane and the control plane. The user plane carries user traffic while the control plane carries signaling in the network. In FIG. **4**, the UPF **418** is in the user plane and all other NFs, i.e., the AMF **404**, SMF **412**, PCF **414**, AF **416**, AUSF **408**, and UDM **410**, are in the control plane. Separating the user and control planes guarantees each plane resource is scaled independently. It also allows UPFs **418** to be deployed separately from control plane functions in a distributed fashion. In this architecture, UPFs **418** may be deployed very close to UEs **400** to shorten the Round Trip Time (RTT) between UEs and the data network for some applications requiring low latency.

The core 5G network architecture is composed of modularized functions. For example, the AMF **404** and SMF **412** are independent functions in the control plane. Separated AMF **404** and SMF **412** allow independent evolution and scaling. Other control plane functions like the PCF **414** and AUSF **408** can be separated as shown in FIG. **4**. Modularized function design enables the 5G core network to support various services flexibly.

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Each NF interacts with another NF directly. It is possible to use intermediate functions to route messages from one NF to another NF. In the control plane, a set of interactions between two NFs is defined as service so that its reuse is possible. This service enables support for modularity. The user plane supports interactions such as forwarding operations between different UPFs **418**.

FIG. **5** illustrates a 5G network architecture **500** using service-based interfaces between the NFs in the control plane, instead of the point-to-point reference points/interfaces used in the 5G network architecture of FIG. **4**. However, the NFs described above with reference to FIG. **4** correspond to the NFs shown in FIG. **5**. The service(s), etc. that a NF provides to other authorized NFs can be exposed to the authorized NFs through the service-based interface. In FIG. **5** the service based interfaces are indicated by the letter "N" followed by the name of the NF, e.g., Namf for the service based interface of the AMF, Nsmf for the service based interface of the SMF, etc. The Network Exposure Function (NEF) **502** and the Network Function (NF) Repository Function (NRF) **504** in FIG. **5** are not shown in FIG. **4** discussed above. However, it should be clarified that all NFs depicted in FIG. **4** can interact with the NEF **502** and the NRF **504** of FIG. **5** as necessary, though not explicitly indicated in FIG. **4**.

Some properties of the NFs shown in FIGS. **4** and **5** may be described in the following manner. The AMF **404** provides UE-based authentication, authorization, mobility management, etc. A UE **400** even using multiple access technologies is basically connected to a single AMF **404** because the AMF **404** is independent of the access technologies. The SMF **412** is responsible for session management and allocates Internet Protocol (IP) addresses to UEs **400**. It also selects and controls the UPF **418** for data transfer. If a UE **400** has multiple sessions, different SMFs **412** may be allocated to each session to manage them individually and possibly provide different functionalities per session. The AF **416** provides information on the packet flow to the PCF **414** responsible for policy control to support Quality of Service (QoS). Based on the information, the PCF **414** determines policies about mobility and session management to make the AMF **404** and SMF **412** operate properly. The AUSF **408** supports authentication function for UEs **400** or similar and thus stores data for authentication of UEs **400** or similar while the UDM **410** stores subscription data of the UE **400**. The DN **420**, not part of the 5G core network, provides Internet access or operator services and similar.

An NF may be implemented either as a network element on a dedicated hardware, as a software instance running on a dedicated hardware, or as a virtualized function instantiated on an appropriate platform, e.g., a cloud infrastructure.

FIG. **6** illustrates an exemplary method for partial access of network information according to some embodiments of the present disclosure. In the embodiment illustrated in FIG. **6**, an Application Function (AF) **416** seeks Access Network (AN) information that is not supported by the AN **402**, resulting in the following steps:

Step **600**. The AF **416** sends a request for AN information to a PCF **414**. In this example, the AF **416** requests timezone information, e.g., for a particular UE **400**.

Step **602**. The PCF **414** forwards the request to an SMF **412**.

Step **604**. The SMF **412** determines that the AN **402** does not support reporting of timezone information. The SMF **412** may know this deficiency in advance, or it may query the AN **402** or a core network node that maintains this information to determine this deficiency.

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Step 606. The SMF 412 notifies the PCF 414 that the AN 402 does not support reporting of timezone information. In some embodiments, the netLocAccSupp attribute is set to "TZR_NOT_SUPPORTED."

Step 608. The PCF 414 forwards that message to the AF 416. In this manner, the AF 416 is notified that the AN 402 does not support reporting timezone information. This can prevent the AF 416 from making false assumptions about the UE's timezone.

The same principle can be applied to any type of AN information that the AF 416 may request, as shown by the next sequence of messages:

Step 610. The AF 416 sends another request to the PCF 414, this time for AN information other than the timezone. This "other AN information" is represented generically as "<AN information>" in FIG. 6.

Step 612. The PCF 414 forwards the request for <AN information> to the SMF 412.

Step 614. The SMF 412 determines that the AN 402 does not support reporting of this specific <AN information>. The SMF 412 may know this deficiency in advance, or it may query the AN 402 or a core network node that maintains this information to determine this deficiency.

Step 616. The SMF 412 notifies the PCF 414 that the AN 402 does not support reporting of <AN information>.

Step 618. The PCF 414 forwards that message to the AF 416. In this manner, the AF 416 is notified that the AN 402 does not support reporting of <AN information>.

FIG. 7 illustrates an exemplary method for partial access of network information according to some embodiments of the present disclosure. In the embodiment illustrated in FIG. 7, an AF 416 seeks AN information that is supported by the AN 402, resulting in the following steps:

Step 700. The AF 416 sends a request for AN information to a PCF 414. In this example, the AF 416 requests timezone information, e.g., for a particular UE 400.

Step 702. The PCF 414 forwards the request to an SMF 412.

Step 704. The SMF 412 determines that the AN 402 supports reporting of timezone information. The SMF 412 may know this functionality in advance, or it may query the AN 402 or a core network node that maintains this information to determine this functionality.

Step 706. The SMF 412 sends a request for timezone information to the AN 402.

Step 708. The SMF 412 receives the timezone information from the AN 402.

Step 710. The SMF 412 forwards the timezone information to the PCF 414, e.g., via the "ueTimeZone" parameter.

Step 712. The PCF 414 forwards the timezone information to the AF 416.

The steps illustrated in FIGS. 6 and 7 are not currently described in existing 3GPP standards. Thus improvements with regards to the current functionality in the interface between the PCF and the SMF that could be made to the existing standards are possible and set forth below:

PCF Request and Report of Access Network Information in the SMF

If the NetLoc functionality is supported by both the SMF 412 and PCF 414, the PCF 414 may request the SMF 412 to report the access network information. If the AN_INFO policy control request trigger is set, upon receiving the "lastReqRuleData" attribute with the "reqData" attribute

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with the value(s) MS_TIME_ZONE and/or USER_LOC_INFO and the "refPccRuleIds" attribute containing the PCC rule identifier(s) corresponding to the PCC rule(s) which is being installed, modified, or removed together, the SMF 412, in an exemplary aspect, shall check if the access network 402 supports the required information:

If the SMF 412 determines that the access network 402 does not support the access network information reporting based on the feature support, the SMF 412 shall immediately inform the PCF 414 by including the "netLocAccSupp" attribute set to "ANR_NOT_SUPPORTED" value in the "UeCampingRep" data structure returned in the "200 OK" response to the policy update notification request.

Otherwise:

- a) If the "reqData" attribute includes the MS_TIME_ZONE value and the SMF 412 determines that the access network 402 does not support the report of the UE time zone, the SMF 412 shall immediately inform the PCF 414 by including the "netLocAccSupp" attribute set to "TZR_NOT_SUPPORTED" value in the "UeCampingRep" data structure returned in the "200 OK" response to the policy update notification request.
- b) If the "reqData" attribute includes:
 - 1) the USER_LOC_INFO value; and/or
 - 2) the MS_TIME_ZONE value and the SMF 412 determines the access network 402 supports the report of UE time zone,
 - 3) then the SMF 412 shall apply appropriate procedures to the EPC access network to obtain the requested access network information as described below.

It should be appreciated that non-support of access network reporting can occur in the interworking scenarios, when the SMF 412 corresponds to an SMF+PGW-C and the access network 402 does not support the reporting of timezone information (Non-3GPP Untrusted Access).

In an exemplary aspect, if access network reporting is supported, the SMF 412 shall apply the Namf_EventExposure service with One-Time Report type as defined in subclause 5.3.1 of 3GPP TS 29.518 if the related information is not available to obtain this information. When the SMF 412 then receives access network information from the AMF 404, the SMF 412 shall provide the required access network information to the PCF 414 by as defined in subclause 4.2.4.1 and set the corresponding attributes as follows:

If the user location information was requested by the PCF 414 and was provided to the SMF 412, the SMF 412 shall provide the user location information within the "userLocationInfo" attribute and the time when it was last known within "userLocationInfoTime" attribute (if available).

If the user location information was requested by the PCF 414 and was not provided to the SMF 412, the SMF 412 shall provide the serving public land mobile network (PLMN) identifier and for standalone non-public network (SNPN) also the network identifier (NID) within the "servingNetwork" attribute.

If the time zone was requested by the PCF 414 and received by the SMF 412, it shall provide it within the "ueTimeZone" attribute.

In addition, the SMF 412 shall provide the AN_INFO policy control request trigger within the "repPolicyCtrlReqTriggers" attribute.

The SMF 412 shall not report any subsequent access network information updates received from the RAN 402

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without any further provisioning or removal of related PCC rules requesting the access network information unless the associated QoS flow or protocol data unit (PDU) session has been released.

PCF Reporting Access Network Information to the AF

This procedure is used by the PCF **414** to report the access network information (i.e., user location and/or user timezone information) to the AF **416** when the “NetLoc” feature is supported.

When the PCF **414** receives the access network information from the SMF **412**, the PCF shall include the “Events-Notification” data type in the body of the HTTP POST request message sent to the AF **416** as described in subclause 4.2.5.2 of TS 29.514. The PCF **414** shall include in the “EventsNotification” data type:

In case of 3GPP access, the user location information in the “eutraLocation” or in the “nrLocation” attribute in the “ueLoc” attribute, if available and required, or in case of untrusted non-3GPP access, the user location information in the “n3gaLocation” attribute in the “ueLoc” attribute, if required, as follows:

- a) the user local IP address in the “ueIpv4Addr” or “ueIpv6Addr” attribute, if available;
- b) the user datagram protocol (UDP) source port in the “portNumber” attribute if available; and
- c) the transmission control protocol (TCP) source port in the “portNumber” attribute if available;

In case of trusted non-3GPP access, the user location information in the “n3gaLocation” attribute in the “ueLoc” attribute, if required, as follows:

- a) the user local IP address in the “ueIpv4Addr” or “ueIpv6Addr” attribute, if available;
- b) the UDP source port in the “portNumber” attribute if available;
- c) the TCP source port in the “portNumber” attribute if available; and
- d) the trusted WLAN access point (TNAP) identifier, that shall consist of:
 - i. the service set identifier (SSID) in the “ssid” attribute;
 - ii. the basic SSID (BSSID) the “bssid” attribute if available; and
 - iii. the TNAP civic address in the “tnapCivicAddress” attribute if available;

the serving PLMN network code and country code in the “plmnId” attribute, if user location information is required but not available in any access;

the UE time zone in the “ueTimeZone” attribute if required and available; and/or

the “netLocAccSupp” attribute set to “TZR_NOT_SUPPORTED” value, if UE timezone information is required but not available in current access.

When the PCF **414** receives notification from the SMF **412** that the access network **402** does not support access network information report, the PCF **414** shall include the “netLocAccSupp” attribute set to “ANR_NOT_SUPPORTED” value in the “EventsNotification” data type in the body of the HTTP POST request message sent to the AF **416** as described in subclause 4.2.5.2 of TS 29.514.

The PCF **414** shall also include an event of the “AfEventNotification” data type in the “evNotifs” attribute with the “event” attribute set to the value “ANI_REPORT.” Note that the PCF **414** receives the access network information from the SMF **412** if it is previously requested by the

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AF **416** or at PDU session termination or at the termination of all the service data flows of the AF session.

The PCF **414** shall not invoke the Npcf_PolicyAuthorization_Notify service operation with the “event” attribute set to the value “ANI_REPORT” to report to the AF **416** any subsequently received access network information, unless the AF **416** sends a new request for access network information.

Example embodiments of at least some aspects of the present disclosure are illustrated in FIGS. **13A-13H** as changes to 3GPP TS 29.512 V15.6.0. Changes are indicated by underlining, strike-throughs, or by being otherwise noted.

Example embodiments of at least some aspects of the present disclosure are illustrated in FIGS. **14A-14I** as changes to 3GPP TS 29.512 V16.3.0. Changes are indicated by underlining, strike-throughs, or by being otherwise noted.

Example embodiments of at least some aspects of the present disclosure are illustrated in FIGS. **15A-15M** as changes to 3GPP TS 29.514 V16.3.0. Changes are indicated by underlining, strike-throughs, or by being otherwise noted.

Example Implementations

FIG. **8** is a schematic block diagram of a network node **800** according to some embodiments of the present disclosure. Optional features are represented by dashed boxes. The network node **800** may be, for example, a radio access node such as a base station **302** or **306** or other node that implements all or part of the functionality described herein. As illustrated, the network node **800** includes a control system **802** that includes one or more processors **804** (e.g., Central Processing Units (CPUs), Application Specific Integrated Circuits (ASICs), Field Programmable Gate Arrays (FPGAs), and/or the like), memory **806**, and a network interface **808**. The one or more processors **804** are also referred to herein as processing circuitry. In addition, the network node **800** may include one or more radio units **810** (if the network node **800** is a radio access node such as a base station **302**) that each includes one or more transmitters **812** and one or more receivers **814** coupled to one or more antennas **816**. The radio units **810** may be referred to or be part of radio interface circuitry. In some embodiments, the radio unit(s) **810** is external to the control system **802** and connected to the control system **802** via, e.g., a wired connection (e.g., an optical cable). However, in some other embodiments, the radio unit(s) **810** and potentially the antenna(s) **816** are integrated together with the control system **802**. The one or more processors **804** operate to provide one or more functions of a network node **800** as described herein. In some embodiments, the function(s) are implemented in software that is stored, e.g., in the memory **806** and executed by the one or more processors **804**.

FIG. **9** is a schematic block diagram that illustrates a virtualized embodiment of the network node **800** according to some embodiments of the present disclosure. This discussion is equally applicable to other types of network nodes. Further, other types of network nodes may have similar virtualized architectures. Again, optional features are represented by dashed boxes.

As used herein, a “virtualized” radio access node is an implementation of the network node **800** in which at least a portion of the functionality of the network node **800** is implemented as a virtual component(s) (e.g., via a virtual machine(s) executing on a physical processing node(s) in a network(s)). As illustrated, in this example, the network node **800** may include the control system **802** and/or the one or more radio units **810** (if the node **800** is a radio access

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node), as described above. The control system **802** may be connected to the radio unit(s) **810** via, for example, an optical cable or the like. The network node **800** includes one or more processing nodes **900** coupled to or included as part of a network(s) **902**. If present, the control system **802** or the radio unit(s) are connected to the processing node(s) **900** via the network **902**. Each processing node **900** includes one or more processors **904** (e.g., CPUs, ASICs, FPGAs, and/or the like), memory **906**, and a network interface **908**.

In this example, functions **910** of the network node **800** described herein are implemented at the one or more processing nodes **900** or distributed across the one or more processing nodes **900** and the control system **802** and/or the radio unit(s) **810** in any desired manner. In some particular embodiments, some or all of the functions **910** of the network node **800** described herein are implemented as virtual components executed by one or more virtual machines implemented in a virtual environment(s) hosted by the processing node(s) **900**. As will be appreciated by one of ordinary skill in the art, additional signaling or communication between the processing node(s) **900** and the control system **802** is used in order to carry out at least some of the desired functions **910**. Notably, in some embodiments, the control system **802** may not be included, in which case the radio unit(s) **810** communicate directly with the processing node(s) **900** via an appropriate network interface(s).

In some embodiments, a computer program including instructions which, when executed by at least one processor, causes the at least one processor to carry out the functionality of network node **800** or a node (e.g., a processing node **900**) implementing one or more of the functions **910** of the network node **800** in a virtual environment according to any of the embodiments described herein is provided. In some embodiments, a carrier comprising the aforementioned computer program product is provided. The carrier is one of an electronic signal, an optical signal, a radio signal, or a computer readable storage medium (e.g., a non-transitory computer readable medium such as memory).

FIG. 10 is a schematic block diagram of the network node **800** according to some other embodiments of the present disclosure. The network node **800** includes one or more modules **1000**, each of which is implemented in software. The module(s) **1000** provide the functionality of network node **800** described herein. This discussion is equally applicable to the processing node **900** of FIG. 9 where the modules **1000** may be implemented at one of the processing nodes **900** or distributed across multiple processing nodes **900** and/or distributed across the processing node(s) **900** and the control system **802**.

Any appropriate steps, methods, features, functions, or benefits disclosed herein may be performed through one or more functional units or modules of one or more virtual apparatuses. Each virtual apparatus may comprise a number of these functional units. These functional units may be implemented via processing circuitry, which may include one or more microprocessor or microcontrollers, as well as other digital hardware, which may include Digital Signal Processor (DSPs), special-purpose digital logic, and the like. The processing circuitry may be configured to execute program code stored in memory, which may include one or several types of memory such as Read Only Memory (ROM), Random Access Memory (RAM), cache memory, flash memory devices, optical storage devices, etc. Program code stored in memory includes program instructions for executing one or more telecommunications and/or data communications protocols as well as instructions for carrying out one or more of the techniques described herein. In

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some implementations, the processing circuitry may be used to cause the respective functional unit to perform corresponding functions according one or more embodiments of the present disclosure.

While processes in the figures may show a particular order of operations performed by certain embodiments of the present disclosure, it should be understood that such order is exemplary (e.g., alternative embodiments may perform the operations in a different order, combine certain operations, overlap certain operations, etc.).

SMF Embodiments

Embodiment 1: A method performed by a Session Management Function, SMF, for providing Access Network, AN, information, the method comprising:

receiving (**602,612,702**), from a Policy Control Function, PCF, a request for AN information;

determining (**604, 614, 704**) whether the AN does not support reporting of the requested AN information;

upon determining that the AN does not support reporting of the requested AN information, sending (**606, 616**), to the PCF, a notification that the AN does not support reporting of the requested AN information; and

upon determining that the AN does support reporting of the requested AN information:

acquiring (**706,708**) the requested AN information; and sending (**710**) the requested AN information to the PCF.

Embodiment 2: The method of embodiment 1, wherein sending (**606,616**) the notification that the AN does not support the reporting the requested AN information comprises sending a UeCampingRep data structure.

Embodiment 3: The method of embodiment 2, wherein sending the notification that the AN does not support the reporting the requested AN information comprises sending (**616**) a UeCampingRep data structure having a netLocAccSupp attribute set to an ANR_NOT_SUPPORTED value.

Embodiment 4: The method of embodiment 2, wherein the requested AN information comprises User Equipment, UE, location information or timezone information, and wherein sending the notification that the AN does not support reporting the requested AN information comprises sending (**606**) a UeCampingRep data structure having a netLocAccSupp attribute set to an TZR_NOT_SUPPORTED value.

PCF Embodiments

Embodiment 5: A method performed by a Policy Control Function, PCF, for providing Access Network, AN, information, the method comprising:

receive (**600, 610, 700**), from an Application Function, AF, a request for AN information;

forwarding (**602,612,702**) the request to a Session Management Function, SMF;

receiving (**606,616,710**), from the SMF, a response to the request; and

forwarding (**608,618,712**) the response to the AF.

Embodiment 6: The method of embodiment 5, wherein the response from the SMF comprises an indication that the AN does not support reporting the requested AN information.

Embodiment 7: The method of embodiment 6, wherein the response from the SMF comprises a UeCampingRep data structure.

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Embodiment 8: The method of embodiment 7, wherein the UeCampingRep data structure comprises a netLocAccSupp attribute set to an ANR_NOT_SUPPORTED value.

Embodiment 9: The method of embodiment 7, wherein the requested AN information comprises User Equipment, UE, location information or timezone information and wherein the UeCampingRep data structure comprises a netLocAccSupp attribute set to an TZR_NOT_SUPPORTED value.

AF Embodiments

Embodiment 10: A method performed by an Application Function, AF, for requesting Access Network, AN, information, the method comprising:

sending (600, 610, 700), to a Policy Control Function, PCF, a request for AN information; and
receiving (608, 618, 712), from the PCF, a response to the request for AN information, wherein the response includes the requested AN information or includes an indication that AN does not support reporting the requested AN information.

Embodiment 11: The method of embodiment 10, wherein the response from the PCF comprises a UeCampingRep data structure.

Embodiment 12: The method of embodiment 11, wherein the UeCampingRep data structure comprises a netLocAccSupp attribute set to an ANR_NOT_SUPPORTED value.

Embodiment 13: The method of embodiment 10, wherein the requested AN information comprises User Equipment, UE, location information or timezone information and wherein the UeCampingRep data structure comprises a netLocAccSupp attribute set to an TZR_NOT_SUPPORTED value.

Apparatus Embodiments

Embodiment 14: A Session Management Function, SMF, comprising:

processing circuitry configured to perform any of the steps of any of the SMF embodiments; and
power supply circuitry.

Embodiment 15: A Policy Control Function, PCF, comprising:

processing circuitry configured to perform any of the steps of any of the PCF embodiments; and
power supply circuitry.

Embodiment 16: An Application Function, AF, comprising:

processing circuitry configured to perform any of the steps of any of the AF embodiments; and
power supply circuitry.

At least some of the following abbreviations may be used in this disclosure. If there is an inconsistency between abbreviations, preference should be given to how it is used above. If listed multiple times below, the first listing should be preferred over any subsequent listing(s).

3GPP Third Generation Partnership Project

5G Fifth Generation

5GC Fifth Generation Core

5GS Fifth Generation System

AF Application Function

AMF Access and Mobility Management Function

AN Access Network

AP Access Point

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ASIC Application Specific Integrated Circuit

AUSF Authentication Server Function

CPU Central Processing Unit

DL Downlink

DN Data Network

DSP Digital Signal Processor

eNB Enhanced or Evolved Node B

EPS Evolved Packet System

E-UTRA Evolved Universal Terrestrial Radio Access

FPGA Field Programmable Gate Array

gNB New Radio Base Station

gNB-DU New Radio Base Station Distributed Unit

HSS Home Subscriber Server

IoT Internet of Things

IP Internet Protocol

LTE Long Term Evolution

MME Mobility Management Entity

MTC Machine Type Communication

NEF Network Exposure Function

NF Network Function

NPLI Network Provided Location Information

NR New Radio

NRF Network Function Repository Function

NSSF Network Slice Selection Function

OTT Over-the-Top

P-CSCF Proxy-Call Session Control Function

PC Personal Computer

PCC Policy and Charging Control

PCF Policy Control Function

PGW Packet Data Network Gateway

QoS Quality of Service

RAM Random Access Memory

RAN Radio Access Network

ROM Read Only Memory

RRH Remote Radio Head

RTT Round Trip Time

SCEF Service Capability Exposure Function

SMF Session Management Function

UDM Unified Data Management

UE User Equipment

UL Uplink

UPF User Plane Function

Those skilled in the art will recognize improvements and modifications to the embodiments of the present disclosure. All such improvements and modifications are considered within the scope of the concepts disclosed herein.

The invention claimed is:

1. A method performed by a Session Management Function, SMF, for providing Access Network, AN, information including User Equipment, UE, timezone information, the method comprising:

receiving a request for reporting one or more of AN information including reporting of the UE timezone information, from a Policy Control Function, PCF;
determining whether an AN supports reporting of the requested AN information;

upon determining that the AN does not support reporting of the requested AN information, sending, to the PCF, a notification that the AN does not support reporting of the requested AN information, comprising sending a UeCampingRep data structure having a netLocAccSupp attribute set to:

an ANR_NOT_SUPPORTED value indicating reporting the requested AN information is not supported;
or

a TZR_NOT_SUPPORTED value indicating reporting the UE timezone information is not supported; and

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upon determining that the AN does support reporting of the requested AN information:
 acquiring the requested AN information; and
 sending the requested AN information to the PCF.

2. The method of claim 1, wherein receiving the request for reporting one or more of AN information comprises receiving a request for user location information.

3. The method of claim 1 comprising sending the UeCampingRep data structure having the netLocAccSupp attribute set to the ANR_NOT_SUPPORTED value.

4. The method of claim 1 comprising sending the UeCampingRep data structure having the netLocAccSupp attribute set to the TZR_NOT_SUPPORTED value.

5. A method performed by a Policy Control Function, PCF, for providing Access Network, AN, information comprising User Equipment, UE, timezone information, the method comprising:

receiving from an Application Function, AF, a request for AN information for reporting one or more of the AN information including reporting of the UE timezone information;
 forwarding the request to a Session Management Function, SMF;
 receiving, from the SMF, a response to the request comprising an indication that an AN does not support reporting the requested AN information, wherein the response from the SMF comprises a UeCampingRep data structure comprising a netLocAccSupp attribute set to:

an ANR_NOT_SUPPORTED value indicating reporting the requested AN information is not supported;
 or
 a TZR_NOT_SUPPORTED value indicating reporting the UE timezone information is not supported; and
 forwarding the response to the AF.

6. The method of claim 5, wherein the netLocAccSupp attribute is set to the ANR_NOT_SUPPORTED value.

7. The method of claim 5, wherein the netLocAccSupp attribute is set to the TZR_NOT_SUPPORTED value.

8. A method performed by an Application Function, AF, for requesting Access Network, AN, information, the method comprising:

sending a request for AN information comprising User Equipment, UE, timezone information to a Policy Control Function, PCF; and
 receiving, from the PCF, a response to the request for AN information, wherein the response includes the requested AN information or includes an indication that an AN does not support reporting the requested AN information,
 wherein the response from the PCF comprises a UeCampingRep data structure comprising a netLocAccSupp attribute set to:

an ANR_NOT_SUPPORTED value indicating reporting the requested AN information is not supported;
 or
 a TZR_NOT_SUPPORTED value indicating reporting the UE timezone information is not supported.

9. The method of claim 8, wherein, when the response includes the netLocAccSupp attribute set to the ANR_NOT_SUPPORTED value.

10. The method of claim 8, wherein the response includes the netLocAccSupp attribute set to the TZR_NOT_SUPPORTED value.

11. A network node for implementing a Session Management Function, SMF, the network node comprising:

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processing circuitry configured to cause the network node to:

receive, from a Policy Control Function, PCF, a request for reporting one or more of Access Network, AN, information including reporting of User Equipment, UE, timezone information;
 determine whether an AN supports reporting of the requested AN information;
 upon determining that the AN does not support reporting of the requested AN information, send to the PCF, a notification that the AN does not support reporting of the requested AN information, comprising sending a UeCampingRep data structure having a netLocAccSupp attribute set to:

an ANR_NOT_SUPPORTED value indicating reporting the requested AN information is not supported; or
 a TZR_NOT_SUPPORTED value indicating reporting the UE timezone information is not supported; and
 upon determining that the AN does support reporting of the requested AN information:
 acquire the requested AN information; and
 send the requested AN information to the PCF.

12. The network node of claim 11, wherein the netLocAccSupp attribute is set to the ANR_NOT_SUPPORTED value.

13. The network node of claim 11, wherein the netLocAccSupp attribute is set to the TZR_NOT_SUPPORTED value.

14. A network node for implementing a Policy Control Function, PCF, the network node comprising:

processing circuitry configured to cause the network node to:

receive, from an Application Function, AF, a request for reporting one or more of Access Network, AN, information including reporting of User Equipment, UE, timezone information;
 forward the request to a Session Management Function, SMF;
 receive, from the SMF, a response to the request comprising an indication that an AN does not support reporting the requested AN information, wherein the response from the SMF comprises a UeCampingRep data structure comprising a netLocAccSupp attribute set to:

an ANR_NOT_SUPPORTED value indicating reporting the requested AN information is not supported;
 or
 a TZR_NOT_SUPPORTED value indicating reporting the UE timezone information is not supported; and
 forward the response to the AF.

15. The network node of claim 14, wherein the netLocAccSupp attribute is set to the ANR_NOT_SUPPORTED value.

16. The network node of claim 14, wherein the requested AN information further comprises UE location information and wherein the netLocAccSupp attribute is set to the TZR_NOT_SUPPORTED value.

17. A network node for implementing an Application Function, AF, the network node comprising:

processing circuitry configured to cause the network node to:

send, to a Policy Control Function, PCF, a request for Access Network, AN, information comprising User Equipment, UE, timezone information; and

receive, from the PCF, a response to the request for AN information, wherein the response includes the requested AN information or includes an indication that an AN does not support reporting the requested AN information,

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wherein the response from the PCF comprises a UeCampingRep data structure having a netLocAccSupp attribute set to:

an ANR_NOT_SUPPORTED value indicating reporting the requested AN information is not supported; or

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a TZR_NOT_SUPPORTED value indicating reporting the UE timezone information is not supported.

18. The network node of claim **17**, wherein the netLocAccSupp attribute is set to the ANR_NOT_SUPPORTED value.

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19. The network node of claim **17**, wherein the requested AN information comprises UE location information and wherein the netLocAccSupp attribute is set to the TZR_NOT_SUPPORTED value.

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